

SenseTime

SenseNebula AIM M30 AI Module

(V 0.1)

Product Manual

www.sensetime.com



Forward

Overview

This manual is applicable to SenseNebula AIM M30 AI model products, and provides information guidelines on product specifications, installation, use and maintenance for users' reference.

Applicable readers

Product installers, users, and maintenance personnel.

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The product specifications and information mentioned in this manual are for reference only, and the content will be updated at any time without notice.

Revision record

Version	Revision data	Modify the content
V0.1	2023/04/12	First release

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1 Product Introduction

1.1 Product summary

SenseNebula AIM M30 AI module (hereinafter referred to as M30) is an embeddable face recognition module product. Based on self-developed living body algorithm and face recognition algorithm from SenseTime, it provides application functions such as living body detection, face recognition and RGB image output, and is widely used in various indoor/semi-outdoor scenes such as office buildings, schools, hospitals, and homes.



1.2 Product Features

Fast and accurate recognition : 0.3s face recognition, recognition accuracy rate is higher than 99.9%, support 0.5-2m recognition distance.

Large capacity of recognition library: Maximum support 20,000 people in the local database.

Powerful detection function: real-time detection and tracking of human faces, accurate detection in scenes such as side faces, half occlusions, and blurs.

Diversified recognition: The device supports recognition of regular faces and faces wearing masks, and user can be recognized normally in the presence of interference such as glasses and beards.

Infrared live detection: effective defense against non-living attacks such as 3D printing, electronic screens, videos, pictures, masks, hoods, etc ;

Interface standardization: the device supports standard USB2.0 interface, which is convenient for external host computer integration.

2 Product specification

2.1 Appearance

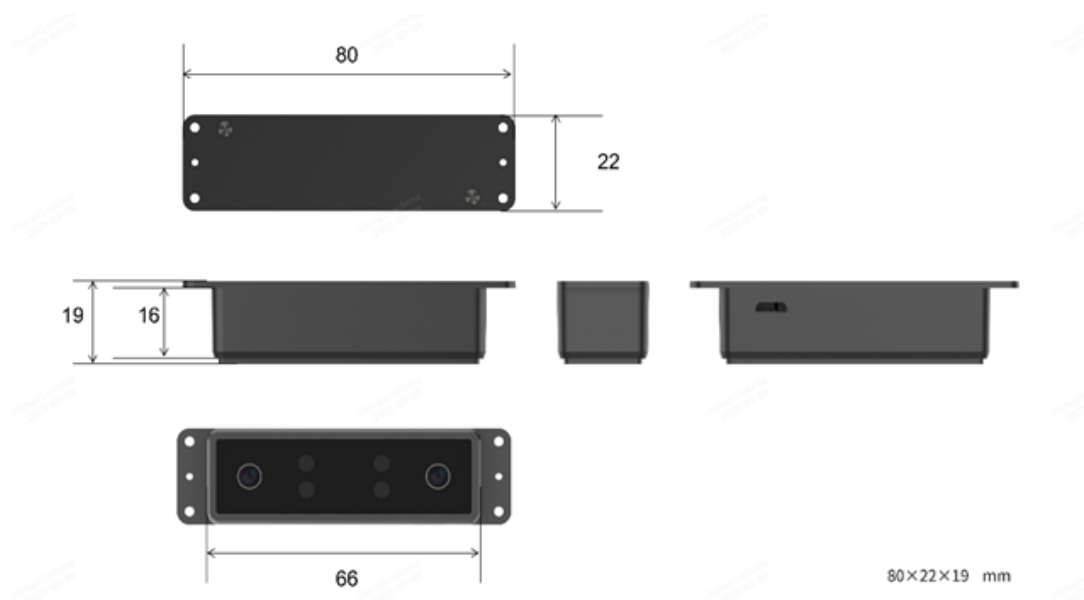


Figure 2-1 Product Appearance

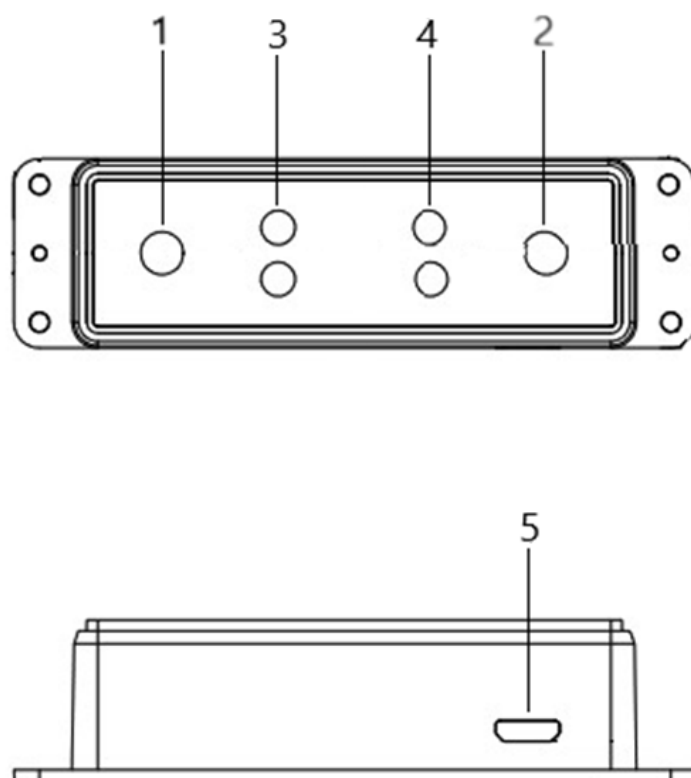
2.2 Specifications

SenseNebula AIM M30 AI module		
Features	Main function	Living body detection, face recognition
	Use environment	Indoor/semi-outdoor
	Recognition distance	0.5-2m(Recommended) 0.3~3m (Max Range)
	Recognition duration	≤0.3s
	Mask Detection	support
	Local face database	20000
	Size	88 X 22 X 19mm
	Weight	50g
	Shell material	Aluminum alloy

	Color	Gray
RGB Camera	Pixels	2 MP Resolution, 1920x1080
	Sensor Size	1/2.8"
	F.NO.	2.8
	FOV	70.4°(D) 62.9°(V)37.9(H)
	Focusing Distance	0.6m
	Focusing Range	0.33m~3.35m
	HDR	Support
IR Camera	Pixels	2 MP Resolution, 1920x1080
	Sensor Size	1/2.8"
	F.NO.	2.8
	FOV	70.4°(D) 62.9°(V)37.9(H)
	Focusing Distance	0.6m
	Focusing Range	0.33m~3.35m
Fill light	Infrared fill light	850nm X4
System	Operating system	Linux
	NPU	1.2T
Power Supply	Through USB port	DC 5V - 1.2A
Interface	Power Switch	Micro USB
	USB interface	Micro USB
	UART Port (inside the shell)	1 UART_TX 2 UART_RX 3 DEBUG_UART_TX 4 DEBUG_UART_RX 5 GND
Environmental requirements	Operation temperature	-20-50°C(Deployment of cooling scheme)
	Storage temperature	-30-80°C
	Operation humidity Storage humidity	0-93% RH

Installation	Installation	Embed
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2.3 Equipment Function & Dimensions



Part number	Function
1	IR camera
2	RGB camera
3&4	Infrared LED
5	Micro USB interface

Figure 2-2 Product Parts (with Shell)

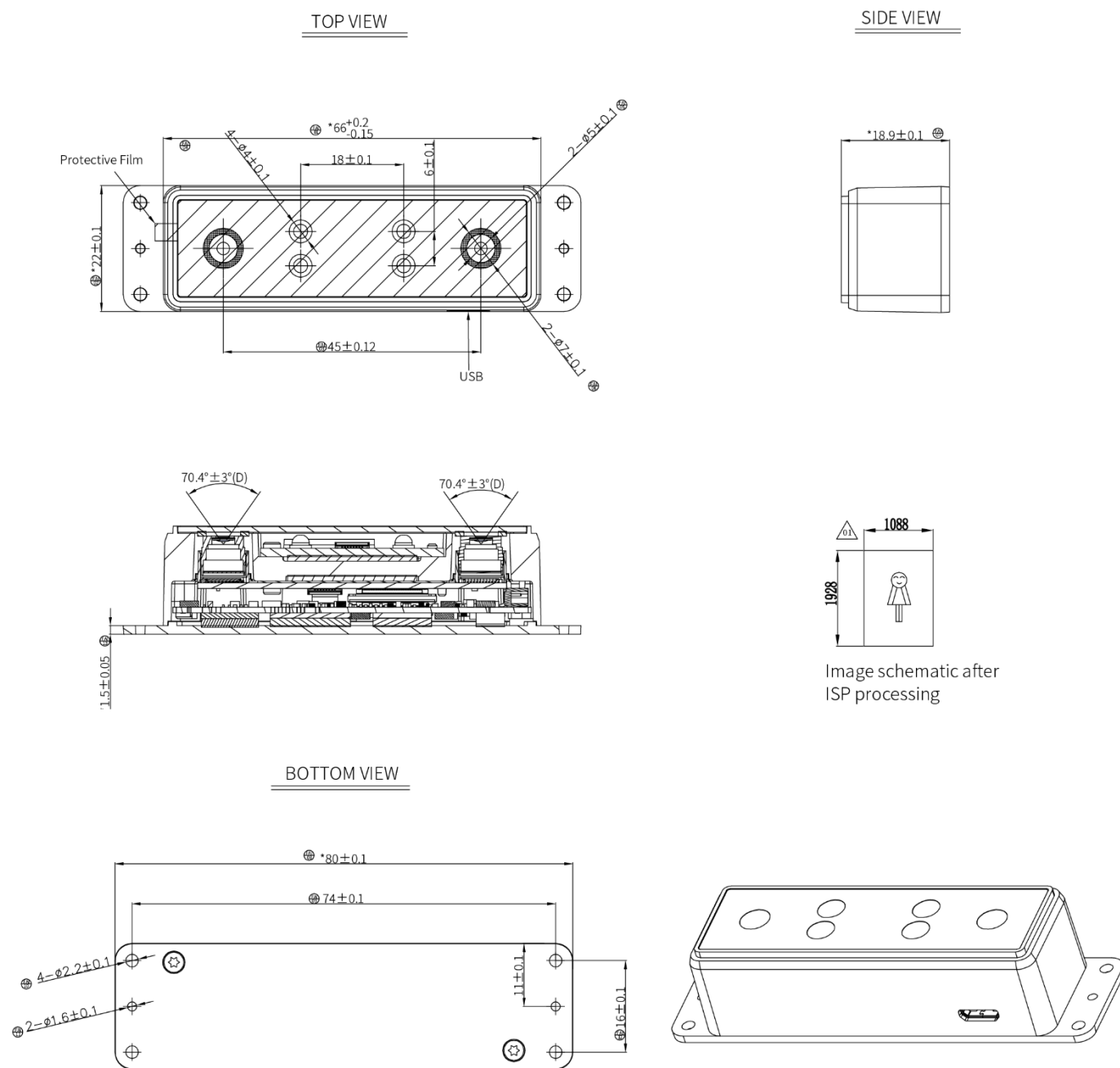
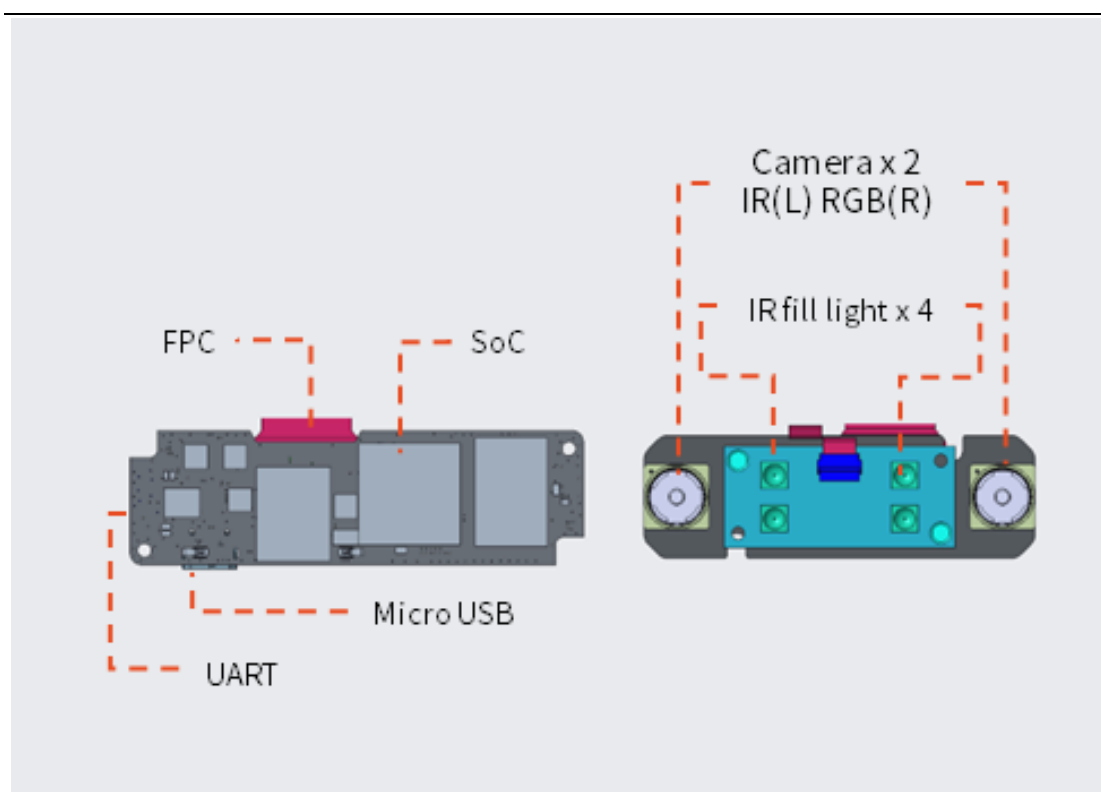


Figure 2-3 Product Dimensions (with Shell)



Function
IR camera
RGB camera
Infrared LED
Micro USB interface
UART interface

Figure 2-4 Product Parts (without Shell)

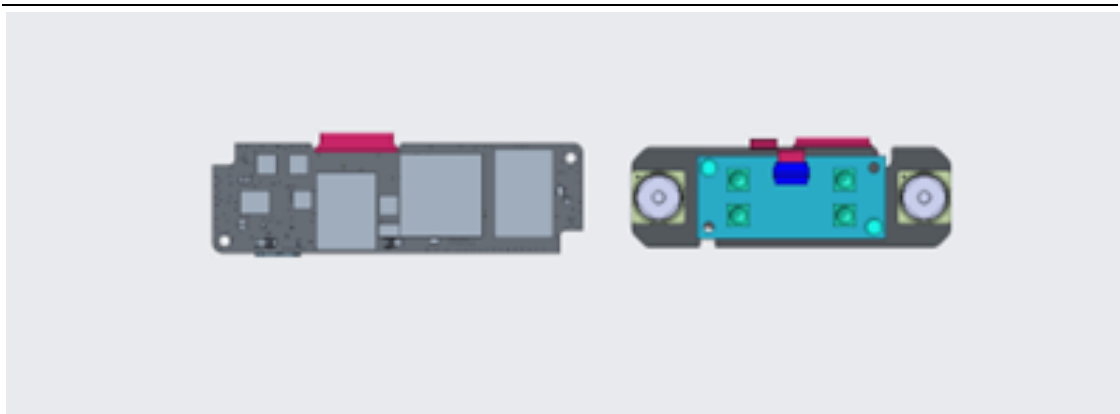


Figure 2-6 Product form A-1 (without Shell, default FPC)

Form A-1 is suitable for devices that require dual cameras on the top, narrow top bezels. The motherboard uses the back shell for heat dissipation.

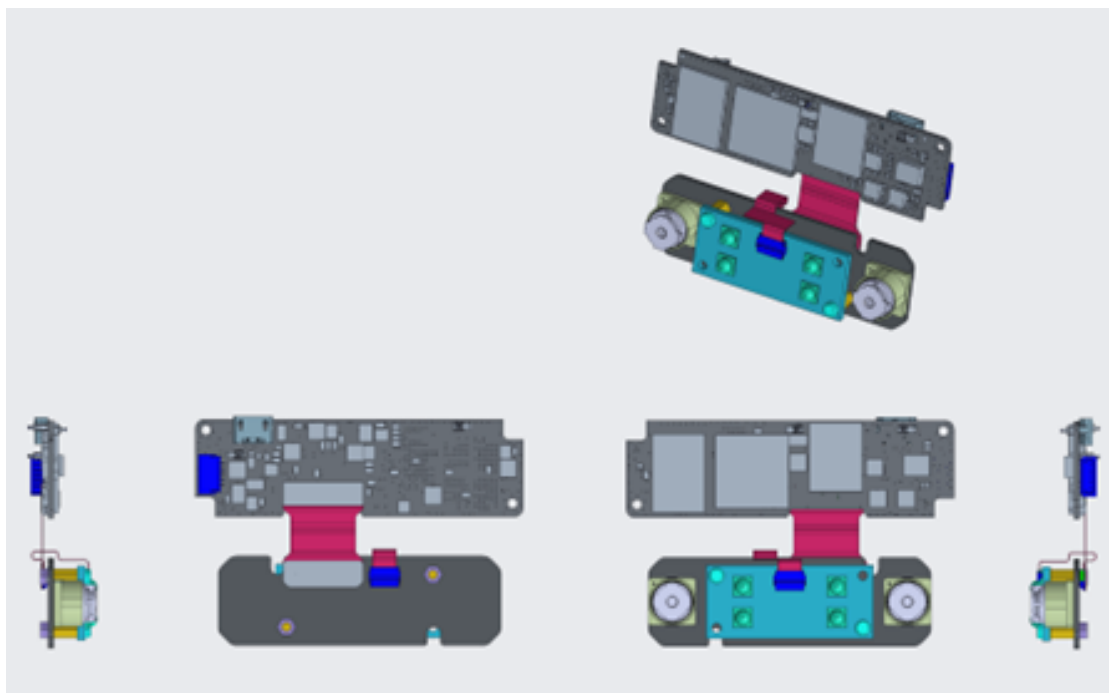


Figure 2-7 Product form A-2 (without Shell, default FPC)

Form A-2 is suitable for devices that require dual cameras on the top, thin bodies. The motherboard uses the front shell or middle frame for heat dissipation.

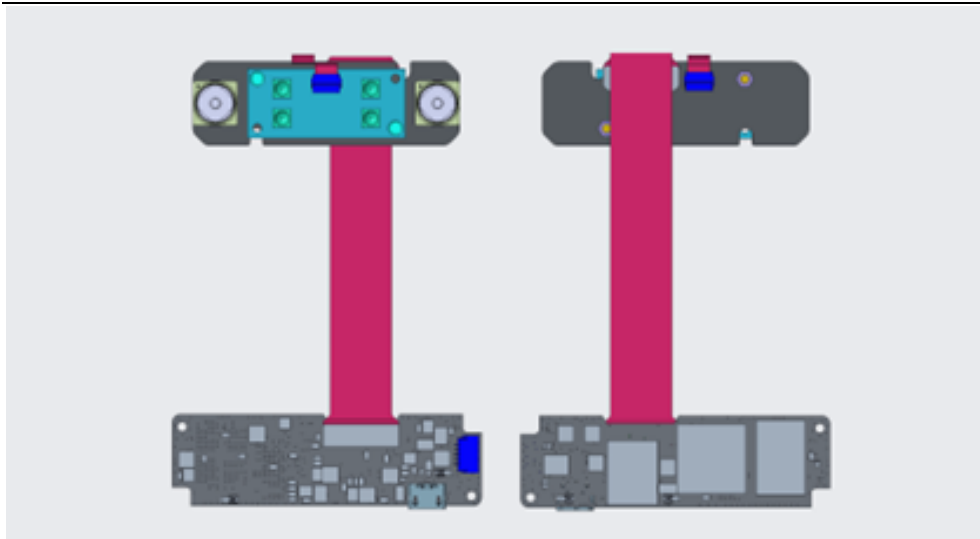


Figure 2-8 Product form B-1 (without Shell, 10cm FPC)

Form B-1 is suitable for devices that require dual cameras on the top, narrow top bezels, and thin bodies. The motherboard uses the back shell for heat dissipation.

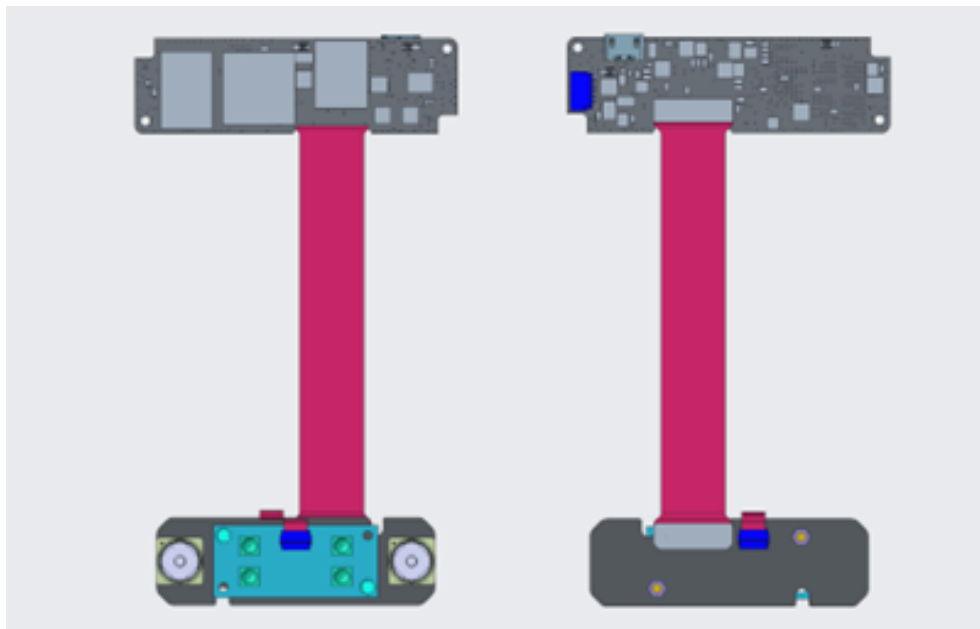


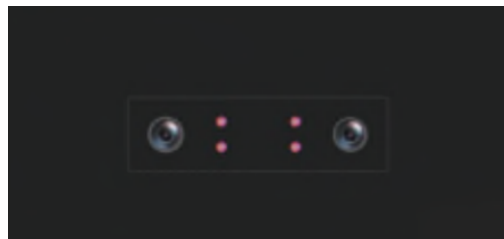
Figure 2-9 Product form B-2 (without Shell, 10cm FPC)

Form B-2 is suitable for devices that require dual cameras on the bottom, with thin and long bodies. The motherboard uses the front shell or middle frame for heat dissipation.

3 Installation tips

The SenseNebula AIM M30 AI module supports only one installation method of embedded installation by default, and other installation methods can be fixed and installed using the M2 screw holes that come with the device.

3.1 Installation effect



◆ Caution :

1. The device software supports the installation direction as shown in the figure below by default. The output video stream only supports vertical screen, and does not support horizontal screen display.
2. The device does not support placement in other directions.



4 Software Operation

4.1 Windows/Ubuntu system

4.1.1 Windows system

1. Software Introduction

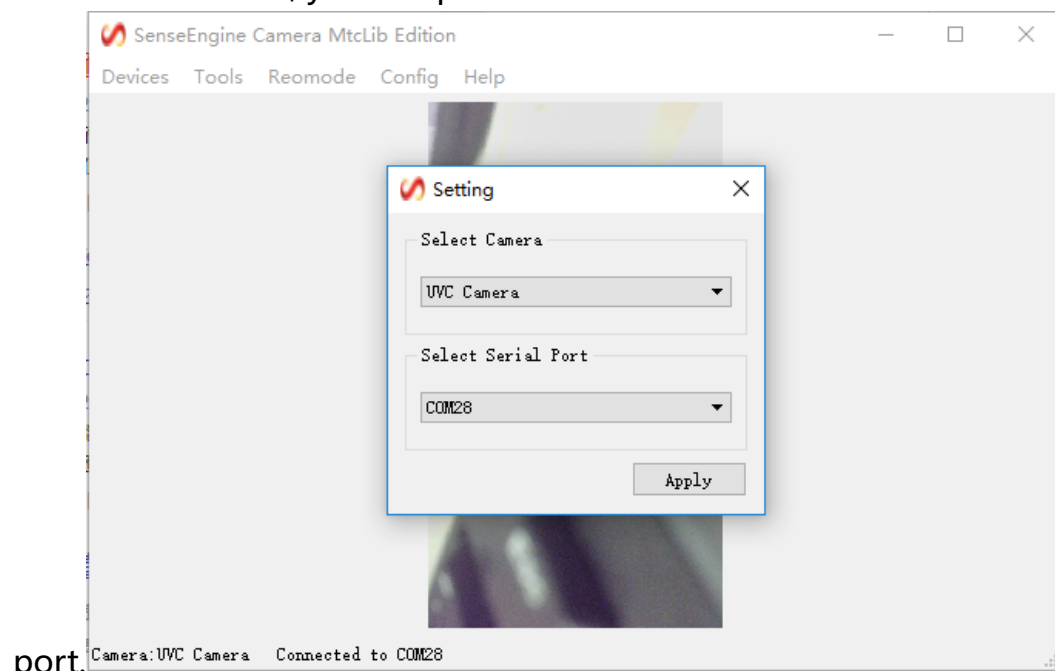
Unzip the package named "libDemouvcCameraWindows_Vx.x.x"
(Based on the latest version), uvcCamera.exe in the directory is the demo software on the windows side.



2. Preparation

(1) Connect the M30 to the PC (minimum win10) device with the Micro USB cable, and wait for the M30 to start up for about 40s. The completion of the start must be based on the appearance of the serial device in the port in the device manager and the appearance of uvcCamera in the camera.

(2) Open the uvcCamera.exe software as shown in the figure, click the Devices ->Setting button, select the correct COM port and camera corresponding to the M30 in the pop-up window, connect to the M30 after confirmation, you can preview the screen and connect to the COM



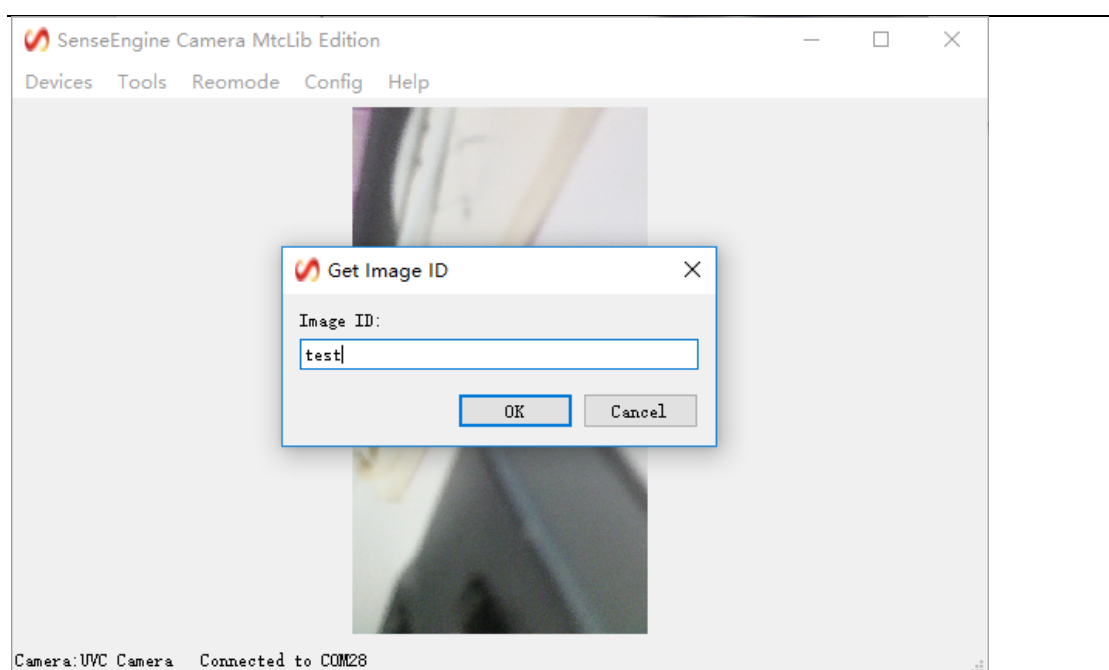
3. Function Description

After the correct connection is set, the functions of the menu bar can be used

1) Take photos and store

Tools->Add base library by camera

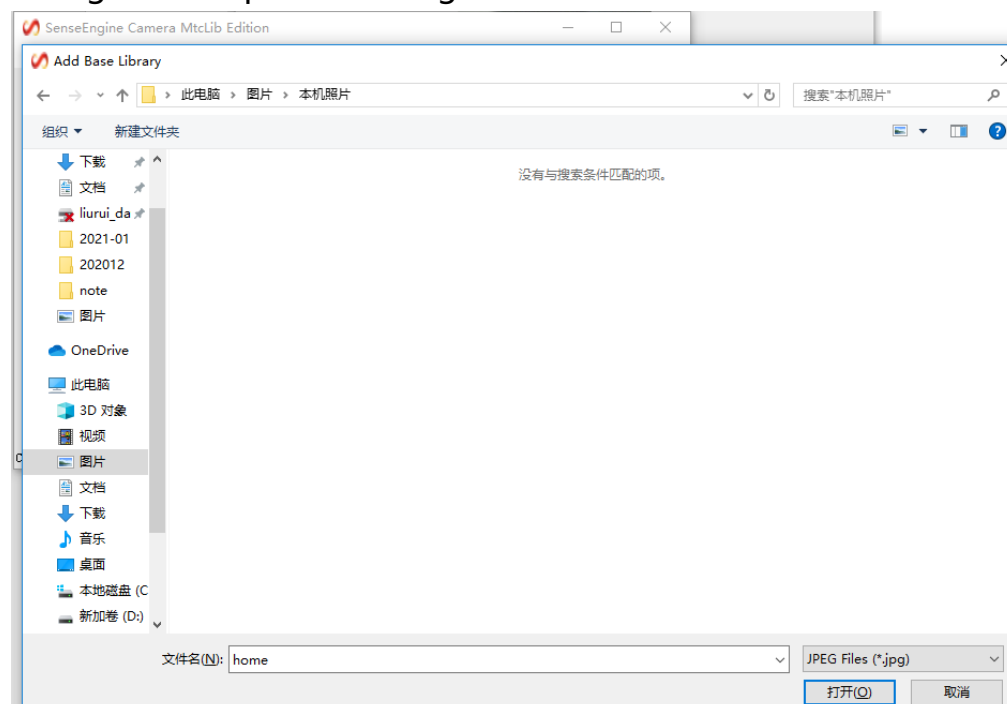
Just enter the ID in the dialog box. At this time, you must ensure that there is a human face in the camera to register successfully.



1) Upload local pictures in the database

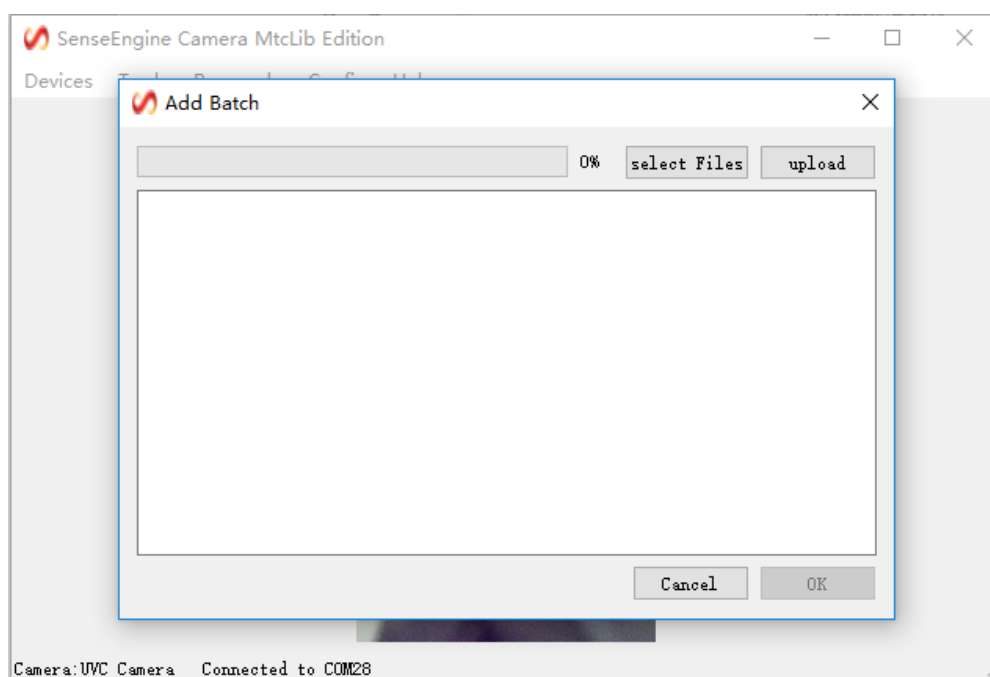
Tools->Add Base Library

After clicking, select the face image on the local computer in the dialog box to upload and register it in the database



2) Batch upload to the library

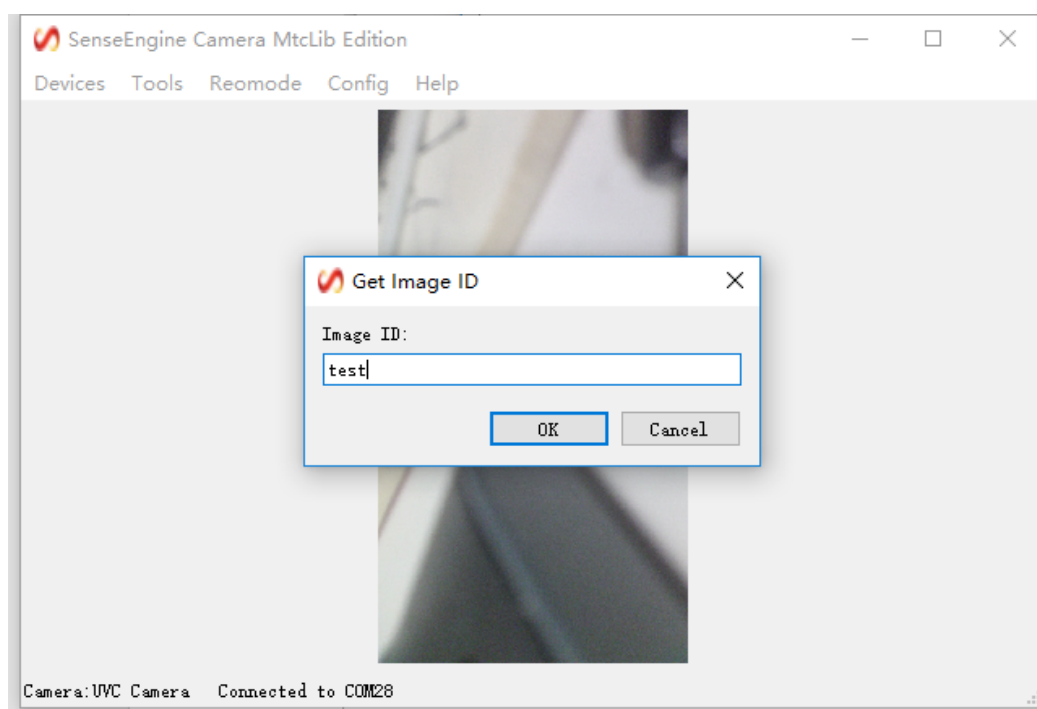
Tools->Batch Addition



In the dialog box, click select Files to select all the basemaps that need to be stored in the library, and click upload to upload

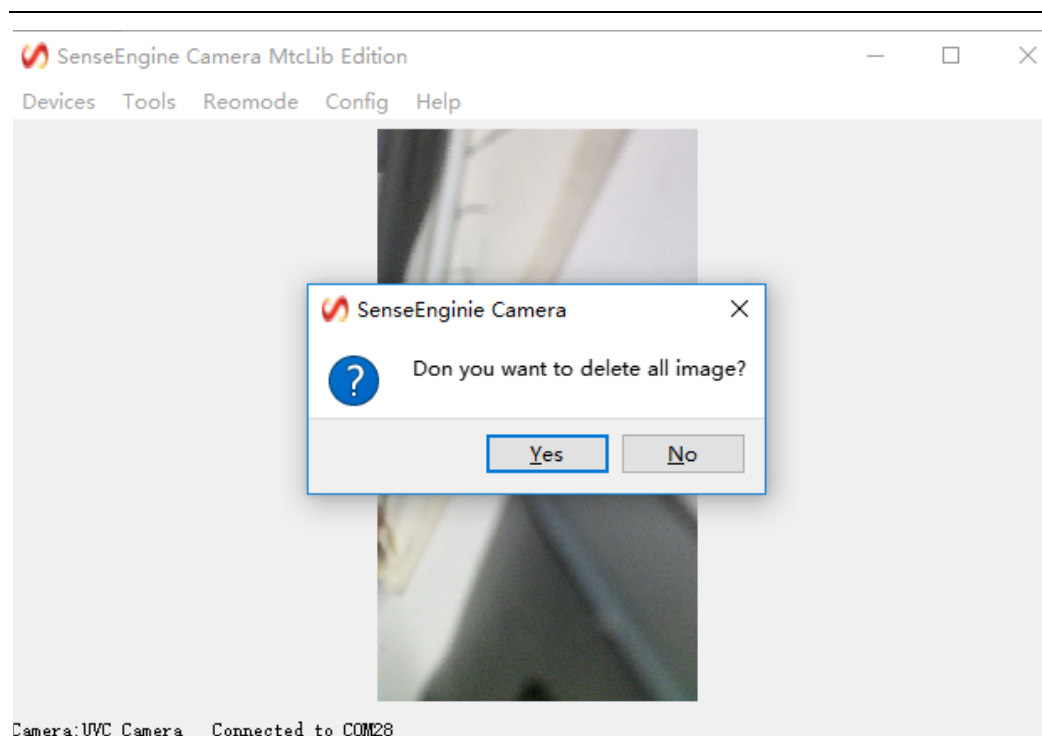
3) Delete the specified ID library personnel

Tools->Delete



4) Empty the library

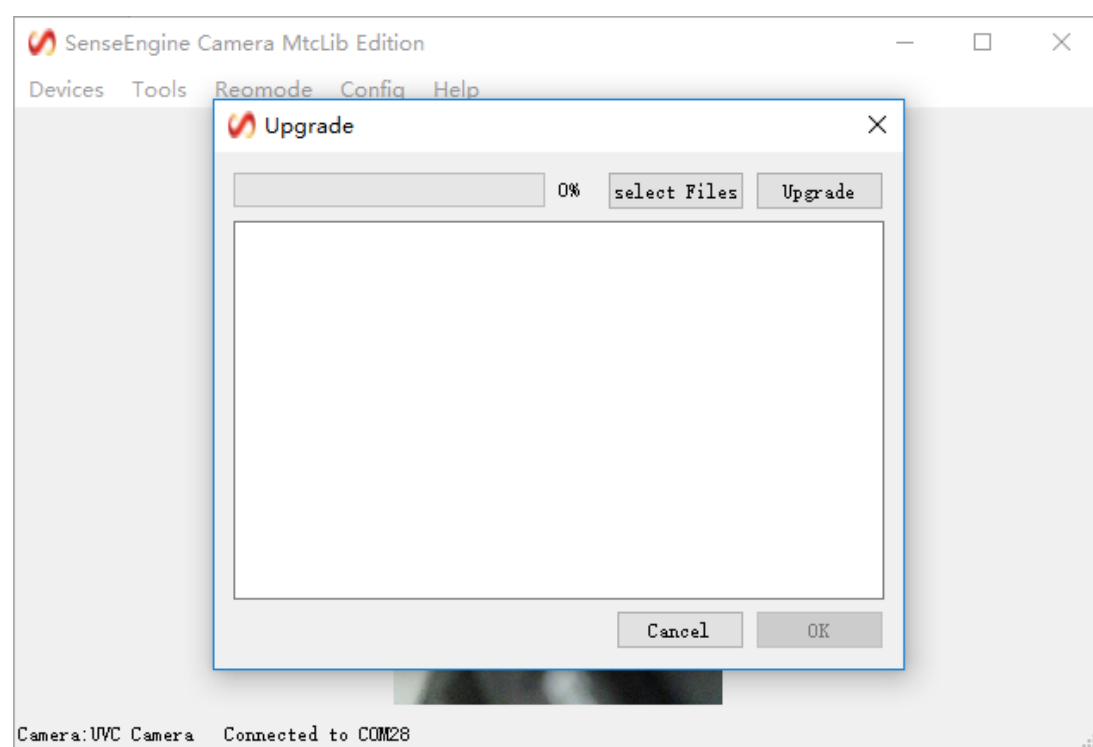
Tools->Delete All



5) Firmware upgrade

Tools->Upgrade

Click Select files in the dialog box, select the upgrade package to upgrade.



6) Open (restore) recognition

Tools->Start Recognize

Recognition will resume after clicking, and the description of the face frame will be restored.

7) Paused recognition

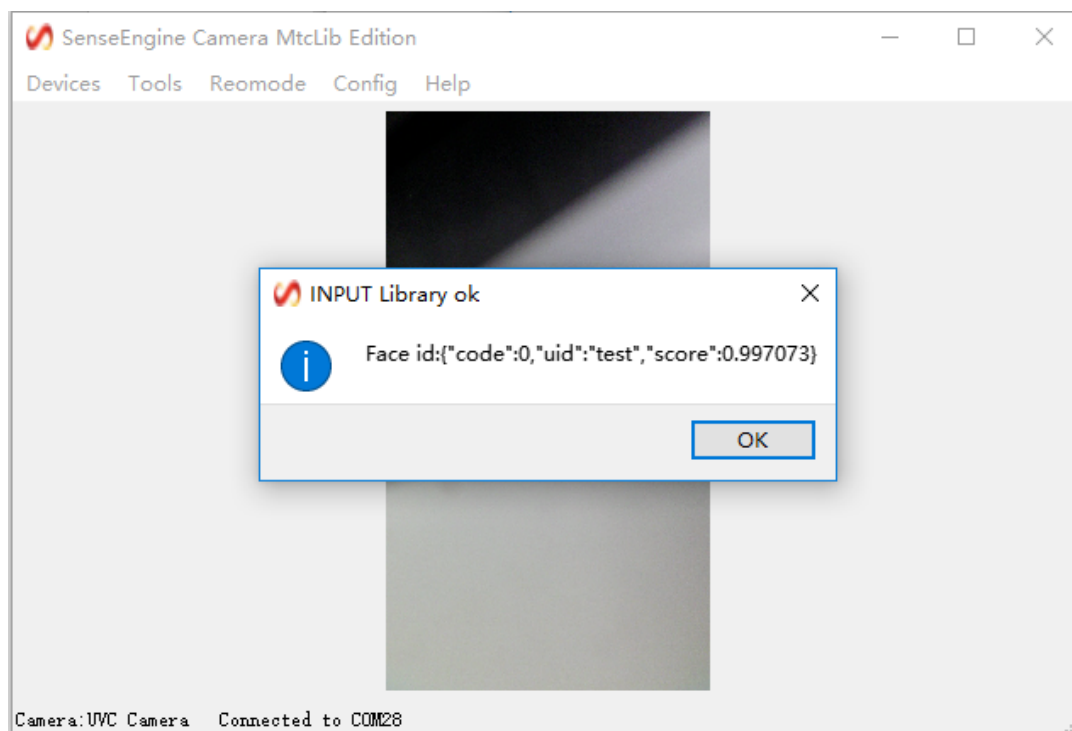
Tools->Stop Recognize

Click to pause recognition and stop the description of the face frame.

8) Static Compare

Tools->Input Library

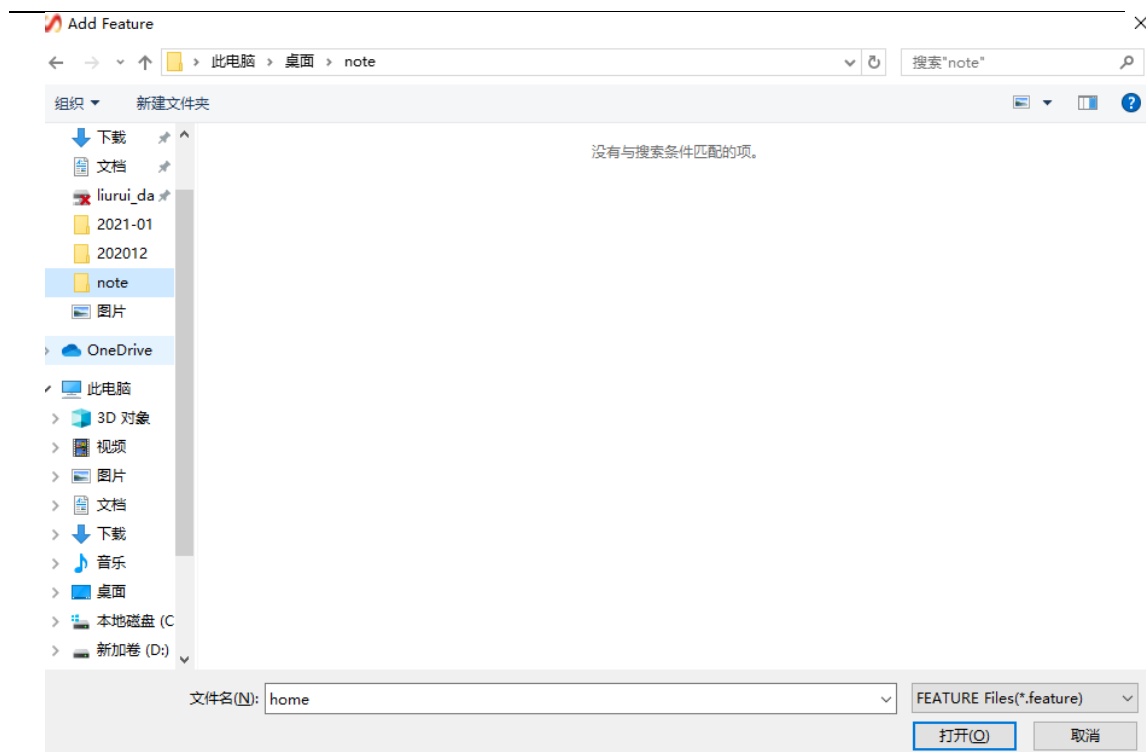
Open the file selection dialog box, and select the picture to be compared on the local computer, pass in a picture, compare it with the person in the library, and return the ID and recognition score.



10) Eigenvalues Upload

Tools->Add feature

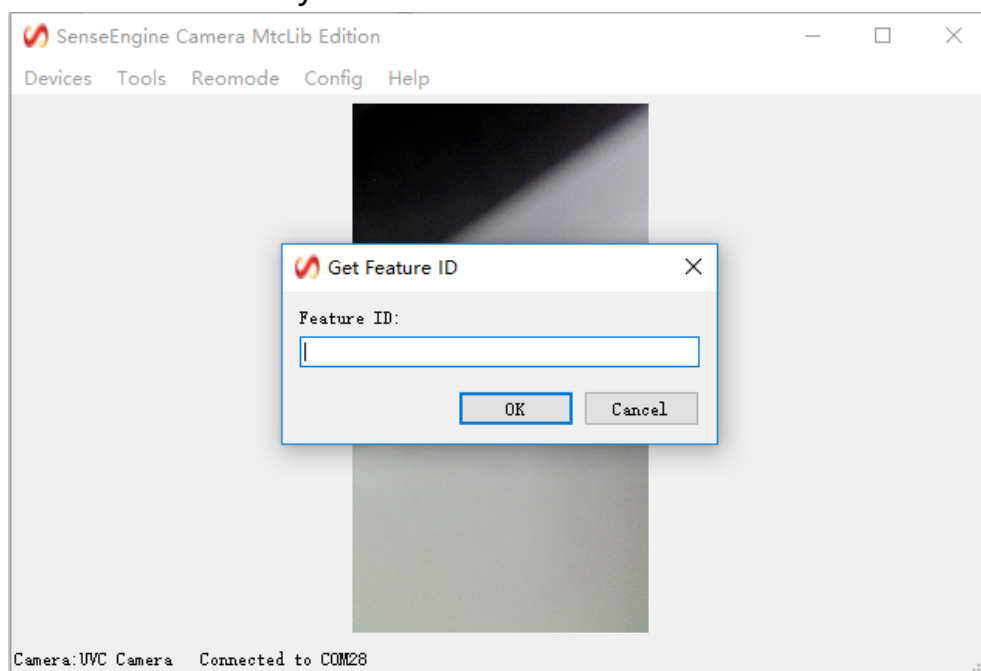
Select the feature value file to enter the ID, and upload eigenvalues.



11) Get the eigenvalue of the specified ID

Tools->Query feature

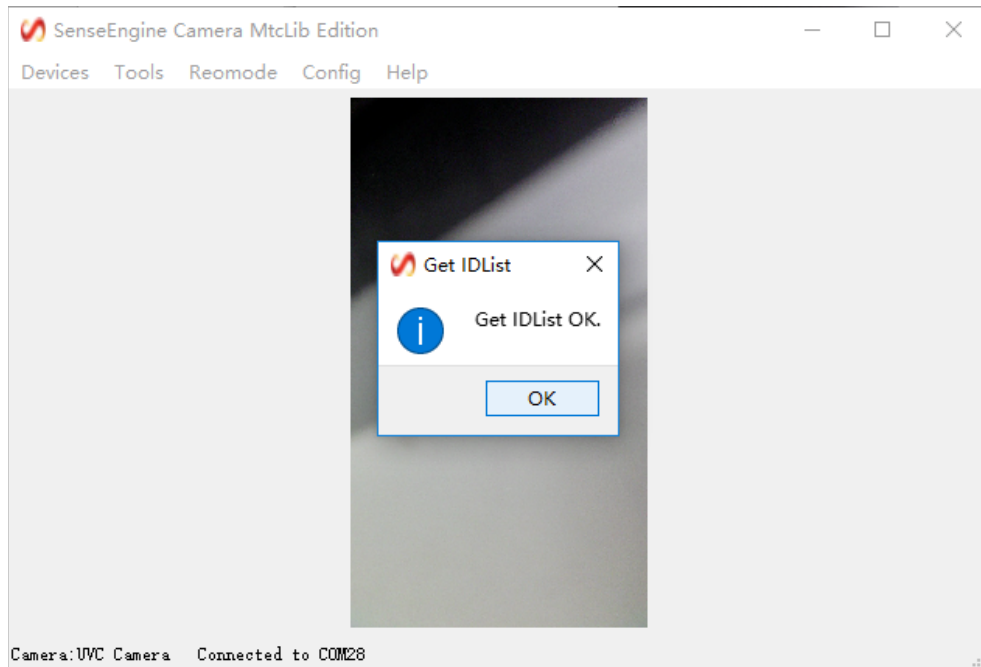
Enter the ID of the registered person to obtain the Eigenvalues of the ID person. The feature value file xxx (ID).feature will be generated in the same level directory



12) Get the ID list

Tools->Get ID list

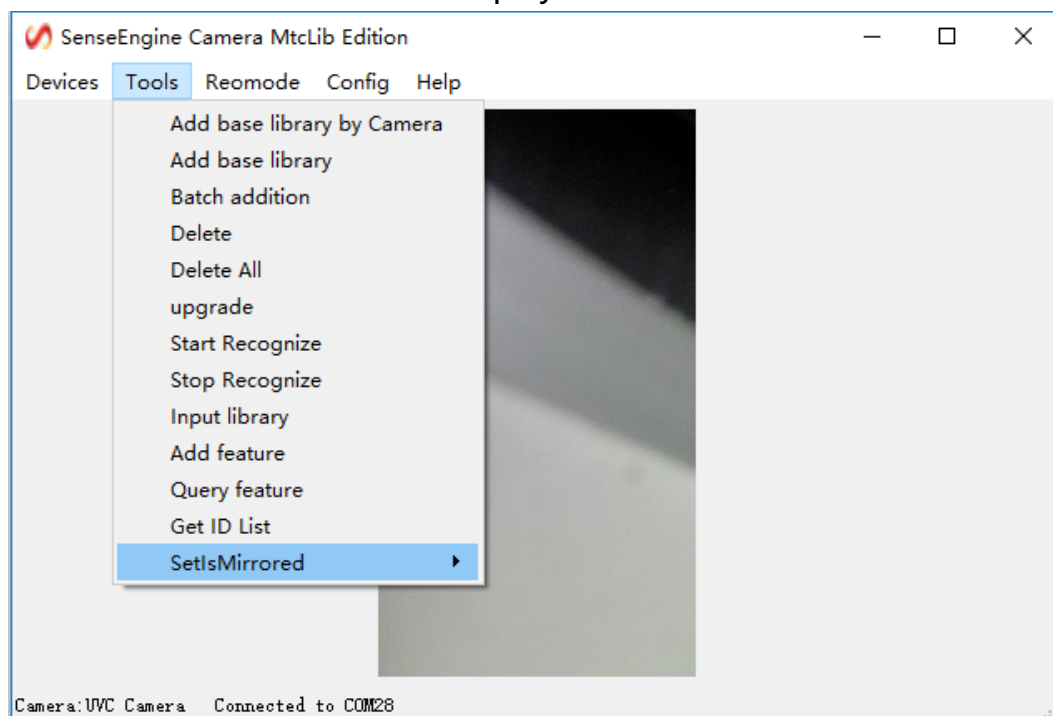
Click to get the ID list of people in the library, the ID List file will be generated in the same level directory of uvcCamera.exe



13) Set screen mirrored

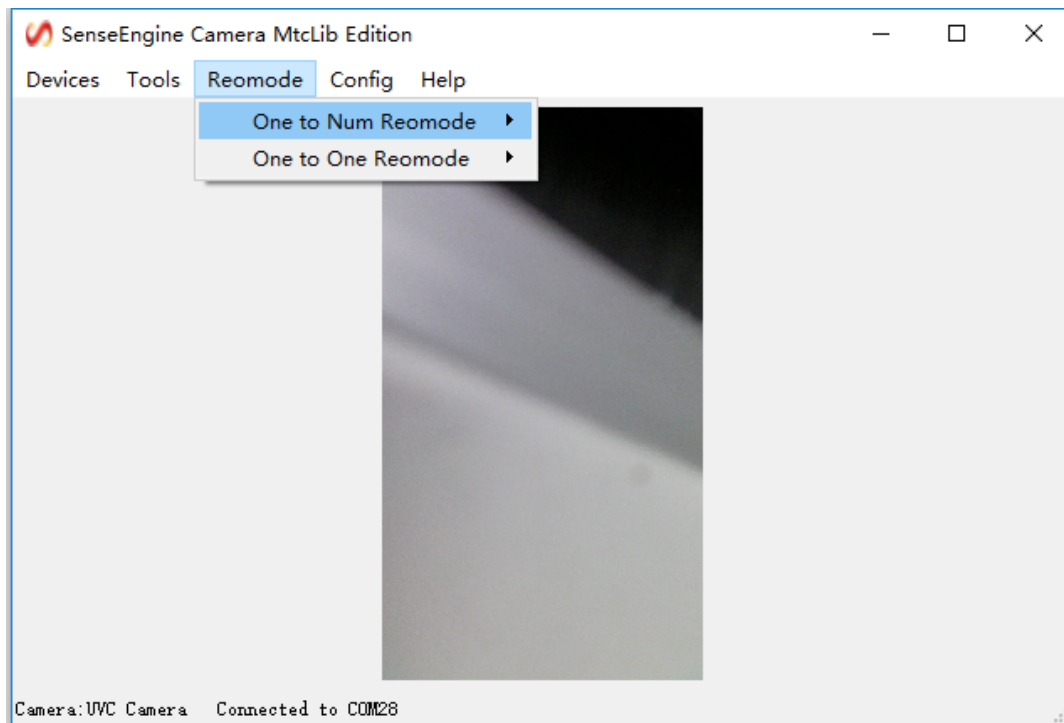
Tools->SetIsMirrored

Set whether the screen is displayed in mirror.



14) Set recognition mode

Click Reomode of menu



One to Num Reomode : 1 : N Recognition Mode

Optional single face or multiple faces, with and without living body recognition

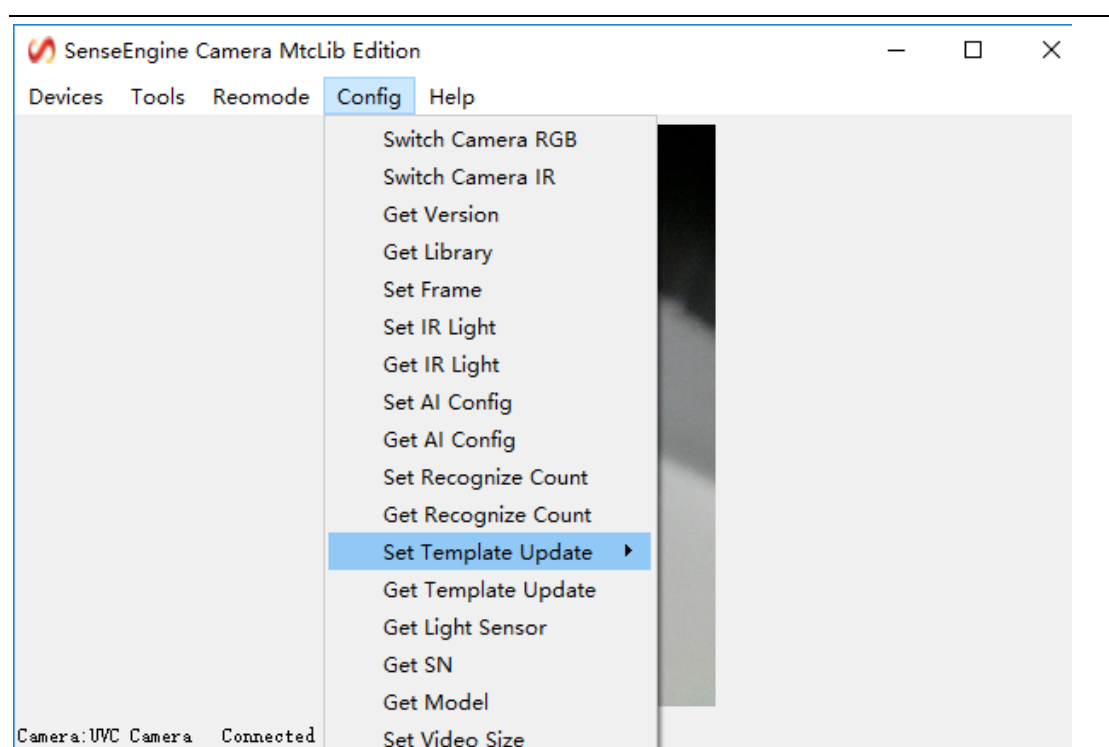
One to One Reomode : 1:1 Recognition Mode

Pass in a picture as the base image and compare it with the face in the video frame

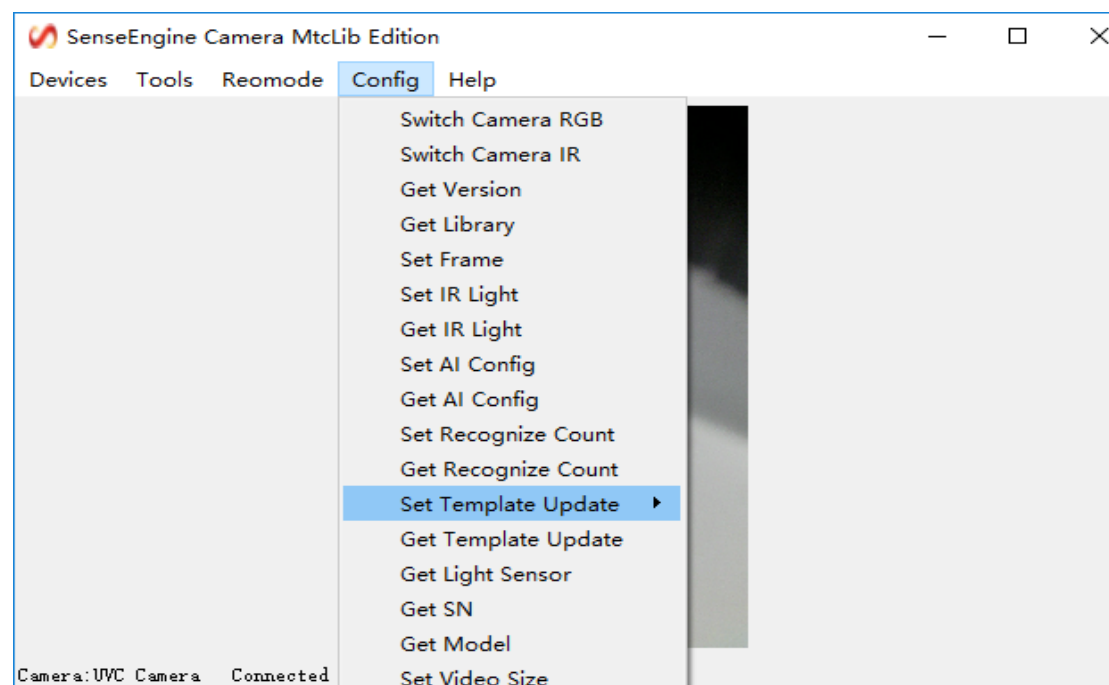
Optional single face or multiple faces, with and without living body recognition

15) Switch RGB Camera Browse

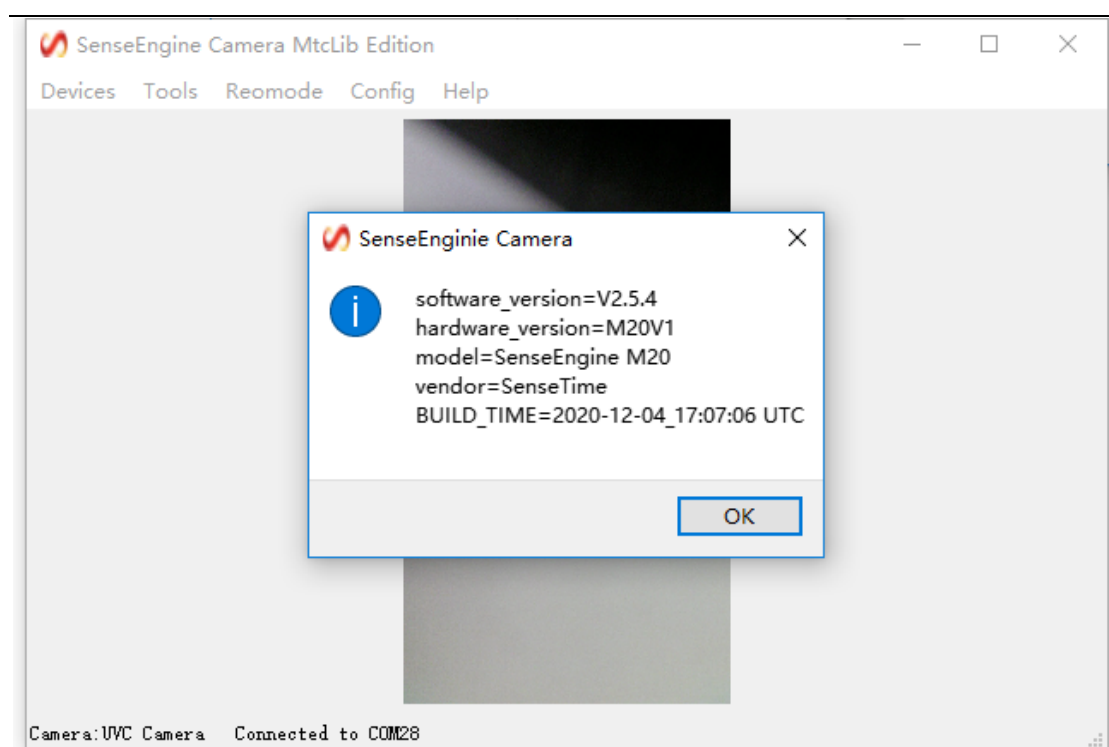
Config->Switch Camera RGB



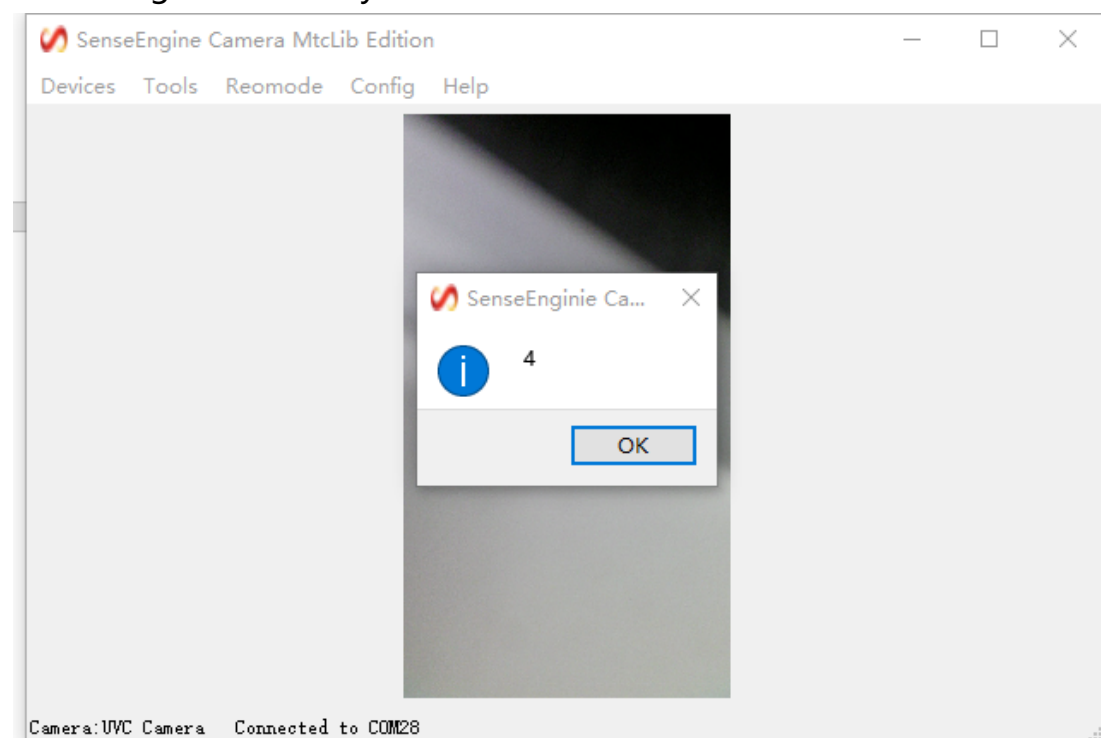
16) Switch IR Camera Browse
Config->Switch Camera IR



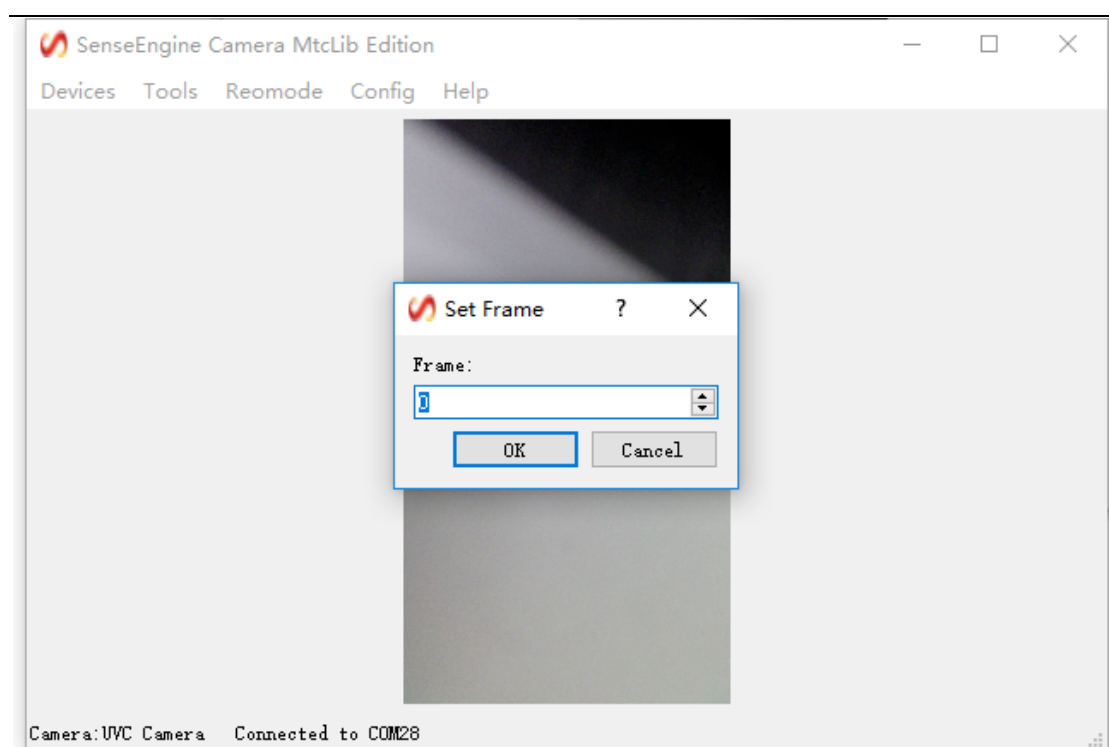
17) Get version information
Config->Get Version



18) Get the quantity of library
Config->Get Library



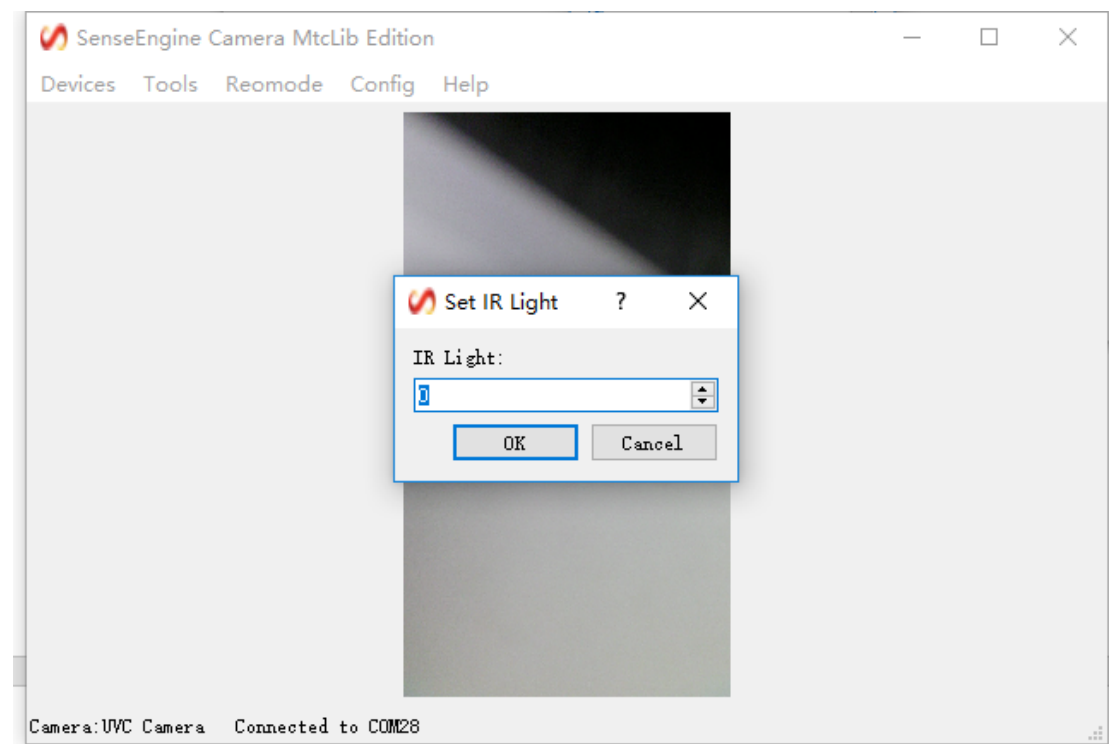
19) Set frame rate
Config->Set Frame,frame range 0 to 25 per second.



20) Setting and get IR LED brightness

Config->Set IR Light, Config->Get IR Light

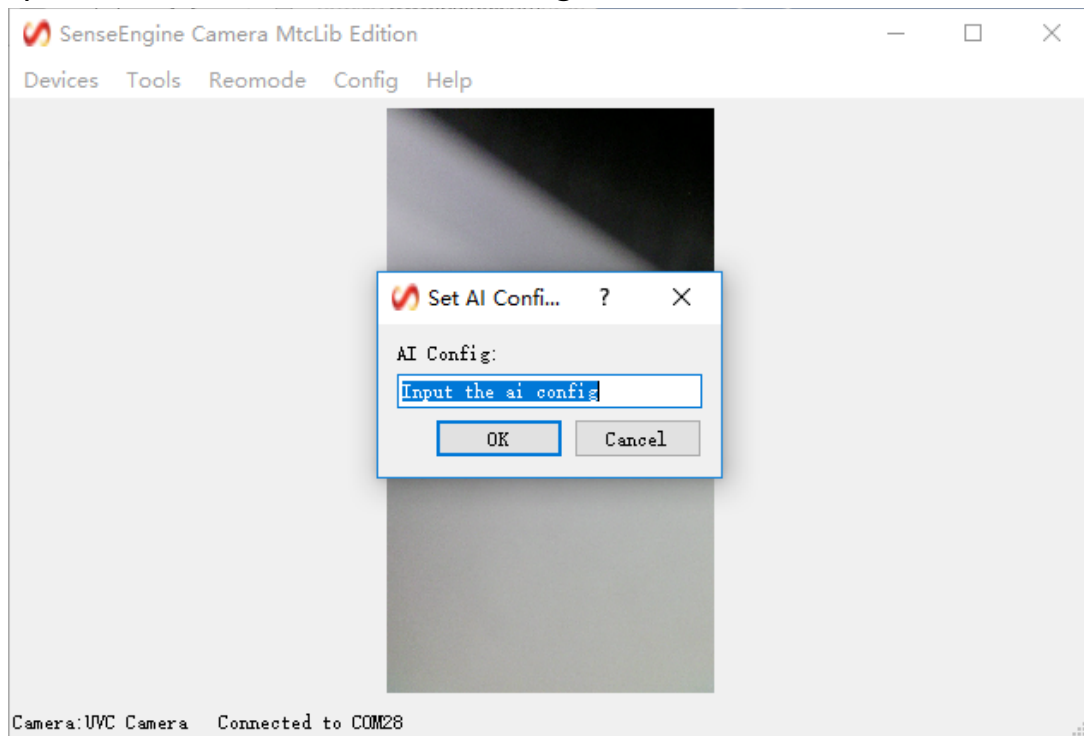
IR LED brightness range 0-200, This value has a positive correlation with M30. fever



21) Set face recognition related parameters

Config->Set AI Config

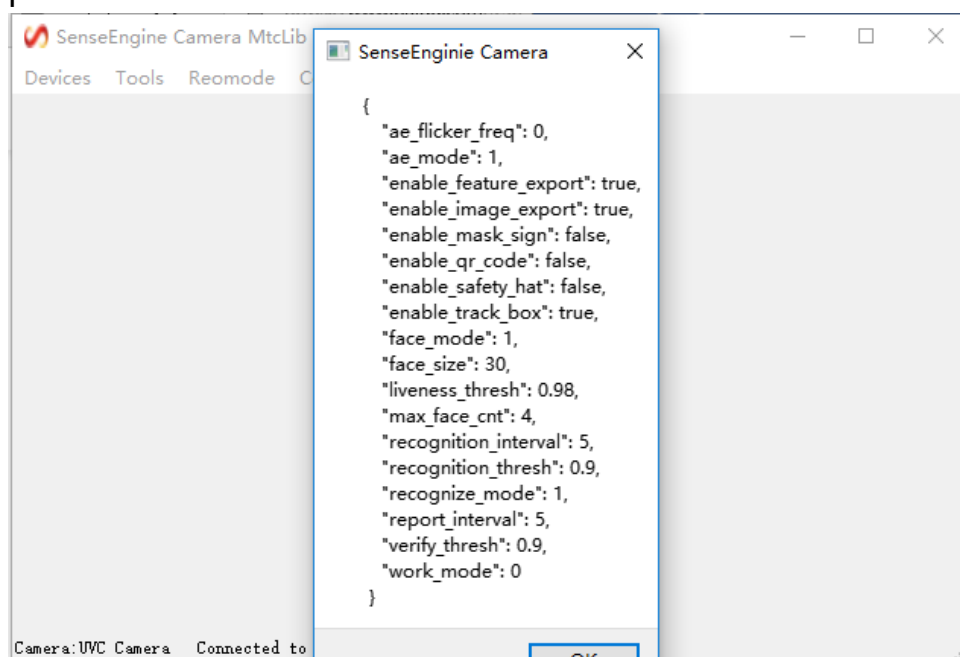
The entire AI config parameter string must be entered, and the specific format can be viewed through Get



22) Get face recognition related parameters

Config->Get AI Config

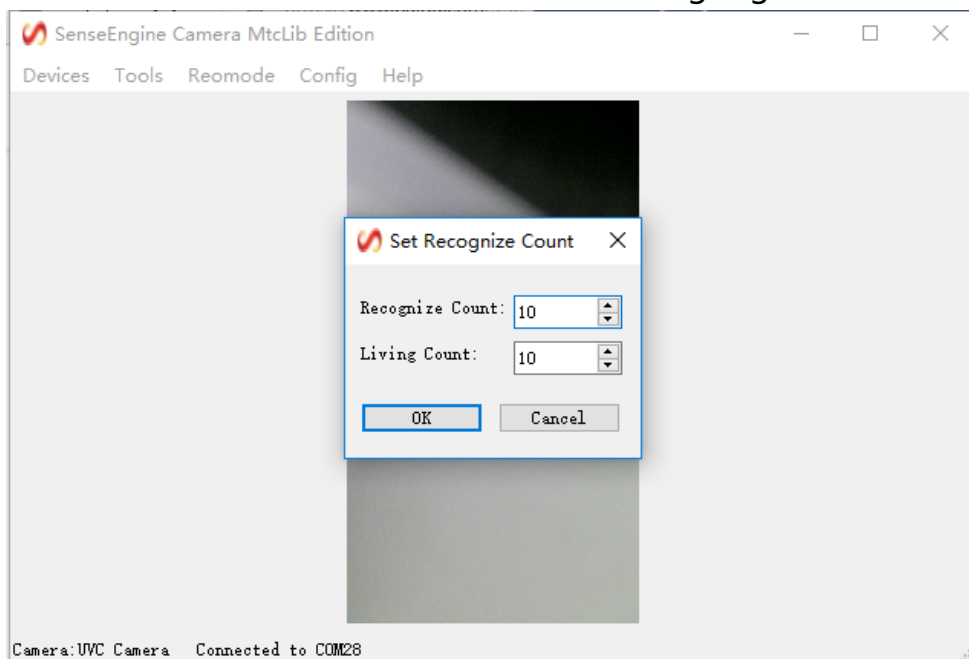
Please refer to the interface documentation for the meaning of the parameters.



23) Set and get recognition times

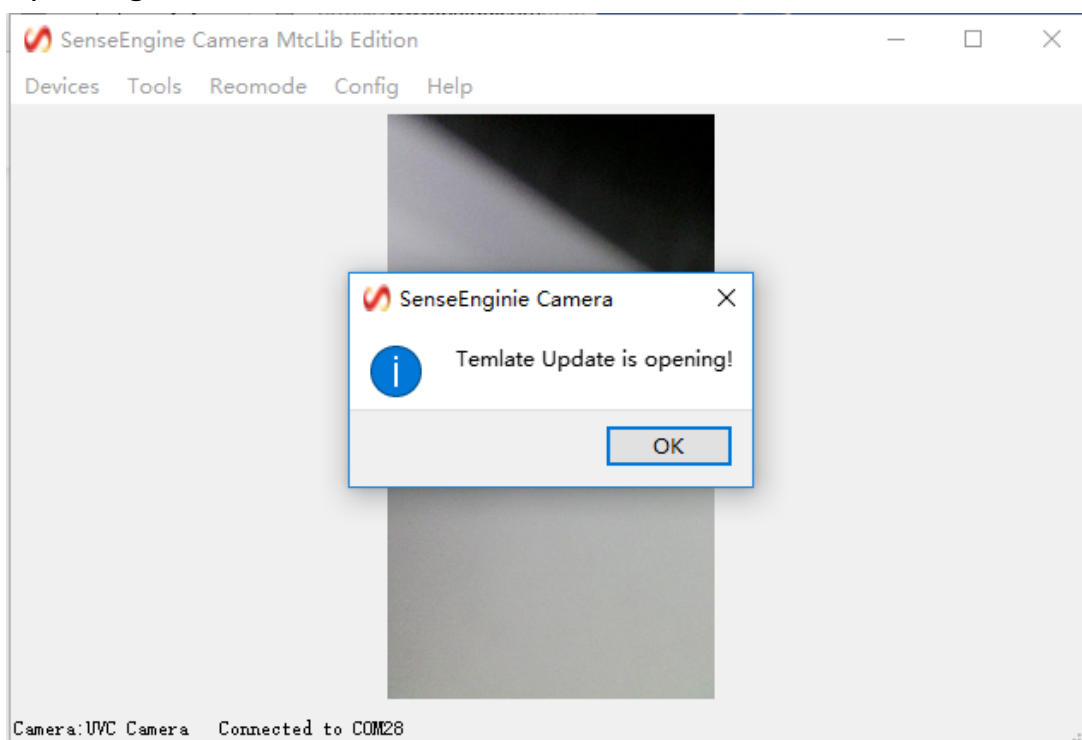
Config->Set Recognize Count, Config->Get Recognize Count

The value range is 0-10, and it is recommended to set the same value for the identification and the number of living organisms



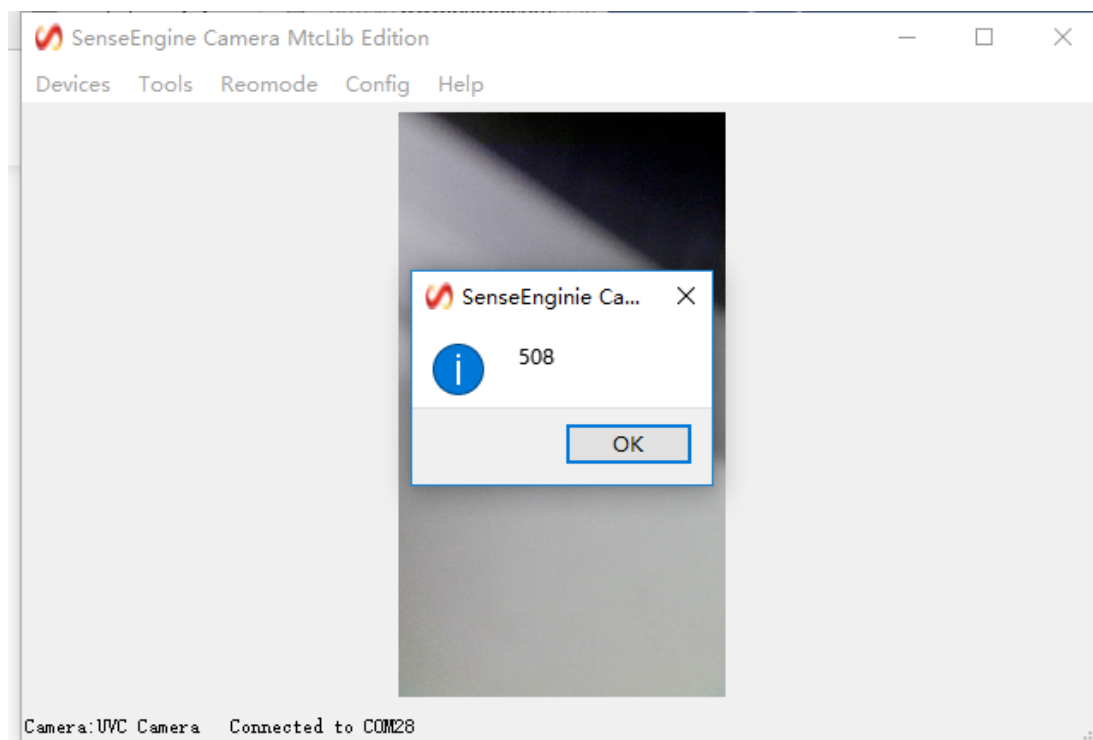
24) Set template update switch and get current status

Config->Set Template Update Set switch, Config->Get Template Update get current status.



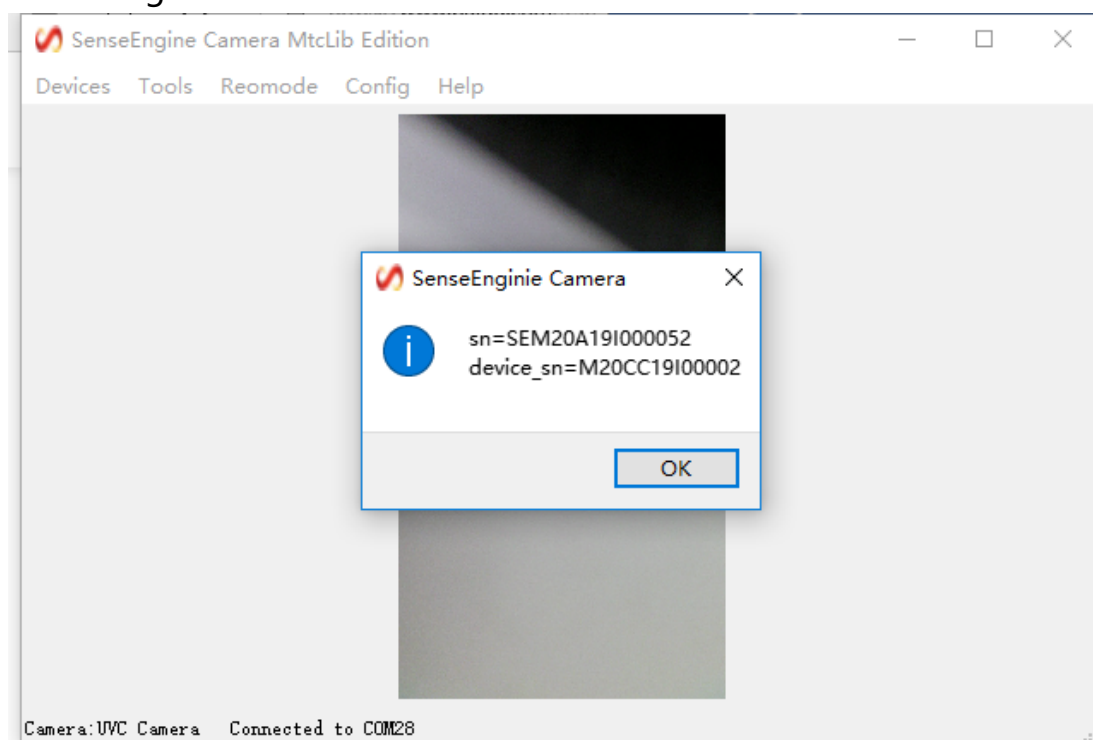
25) Get current light sensor value

Config->Get Light Sensor



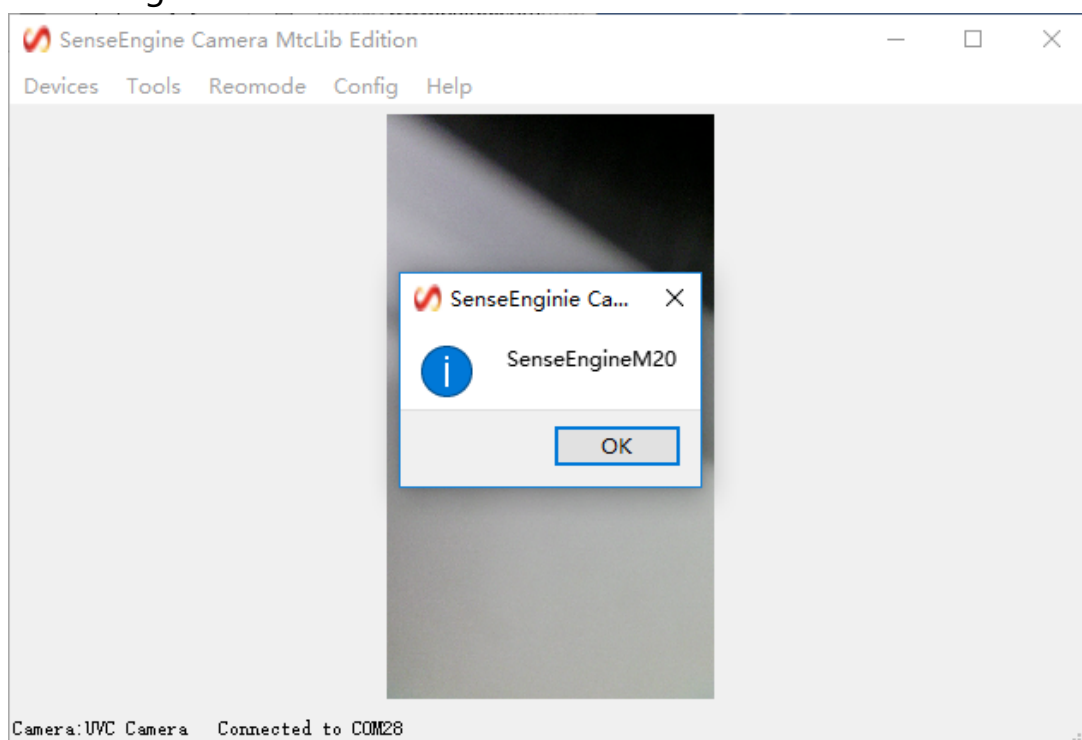
26) Get the SN number

Config->Get SN



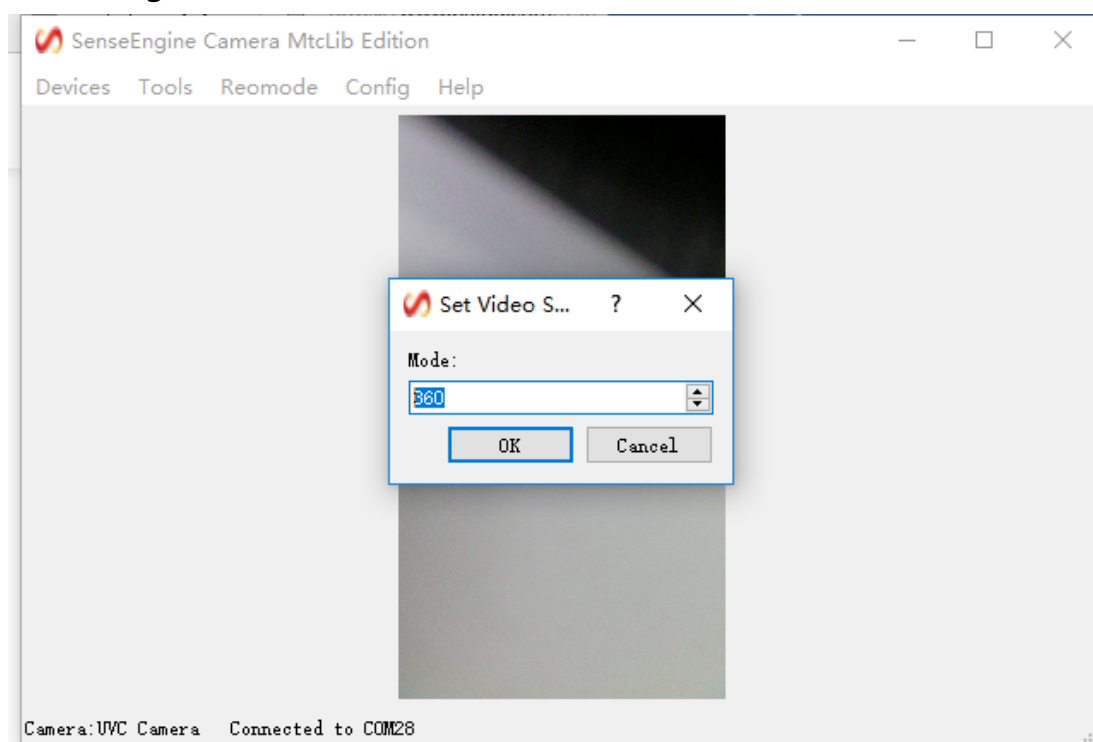
27) Get the product model

Config->Get Model



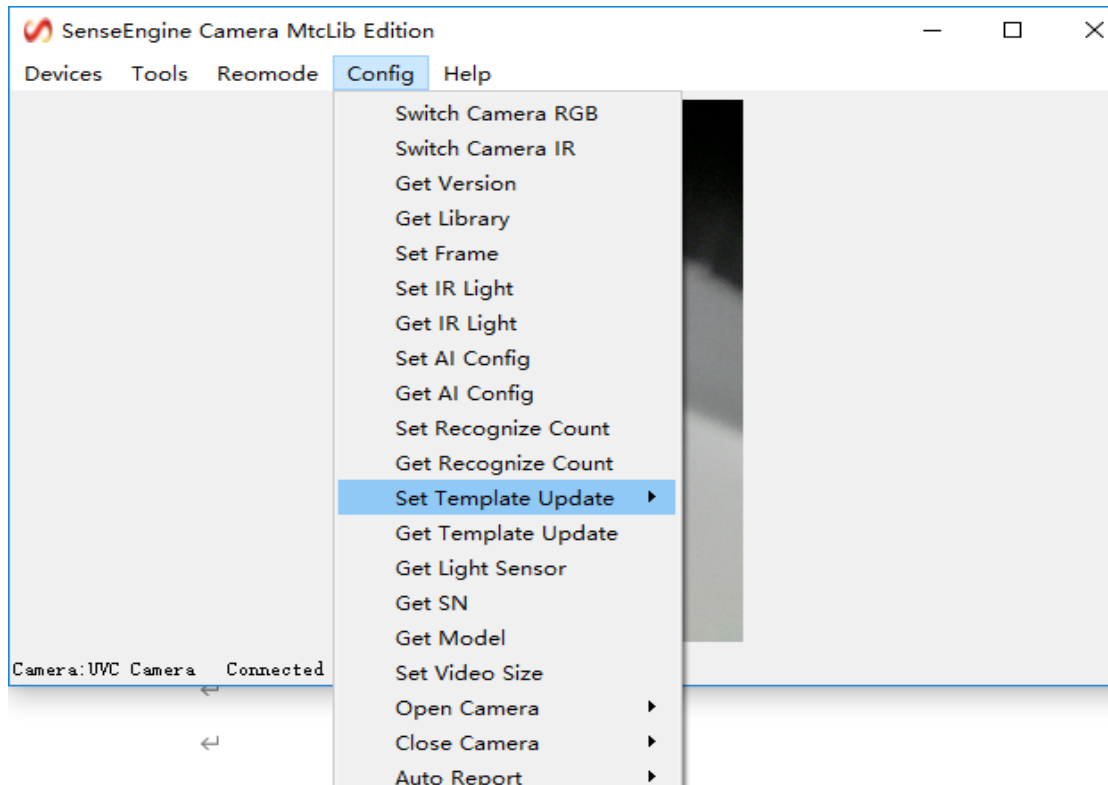
28) Set screen size

Config->Set Screen Size



29) Switch camera

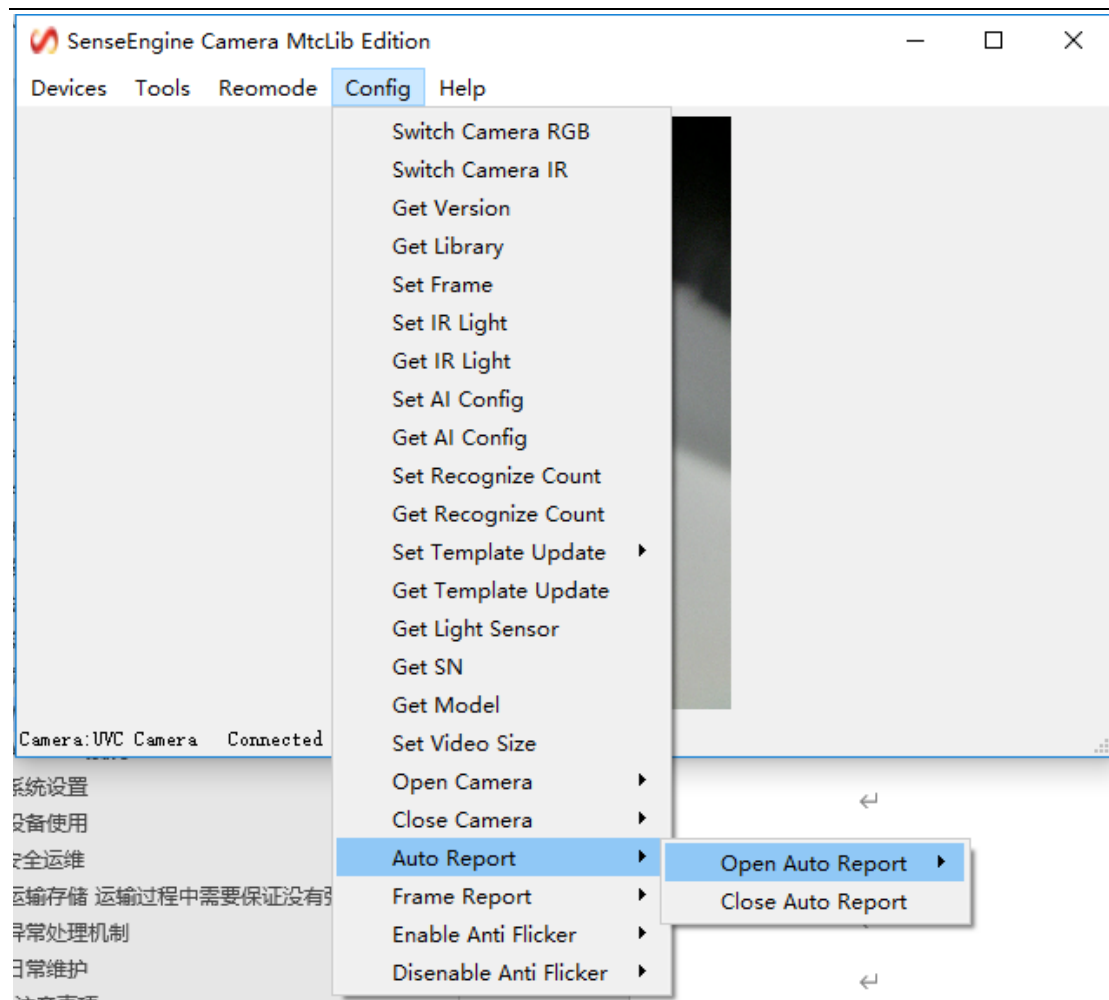
Config->Open Camera, Config->Close Camera



30) Active report switch

Config->Auto Report

You can select active reporting in the submenu, and the types of images to be reported, including: RGB background image, RGB face image, and IR image.



31) Frame report upload switch

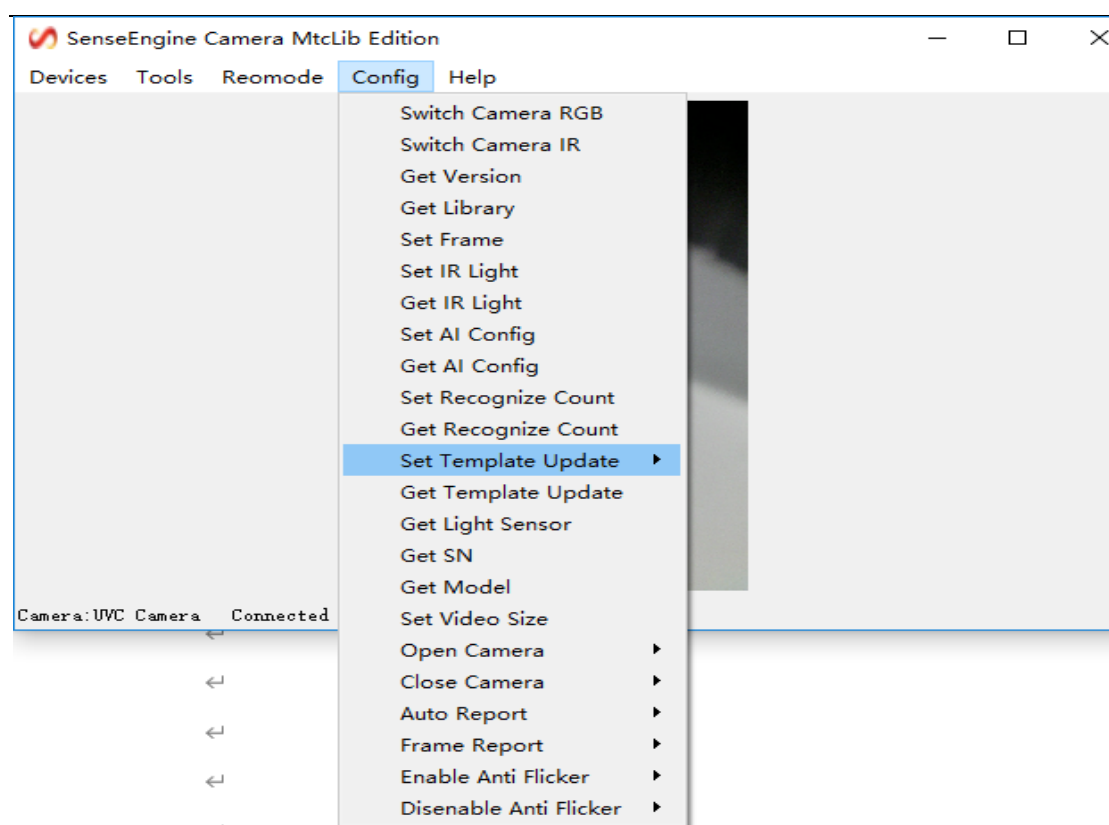
Config->Frame Report

Turning on this switch means that the structured data of face information will be inserted in the UVC frame data and reported together.

32) Anti flicker

Config->Enable anti flicker, Config->Disable anti flicker

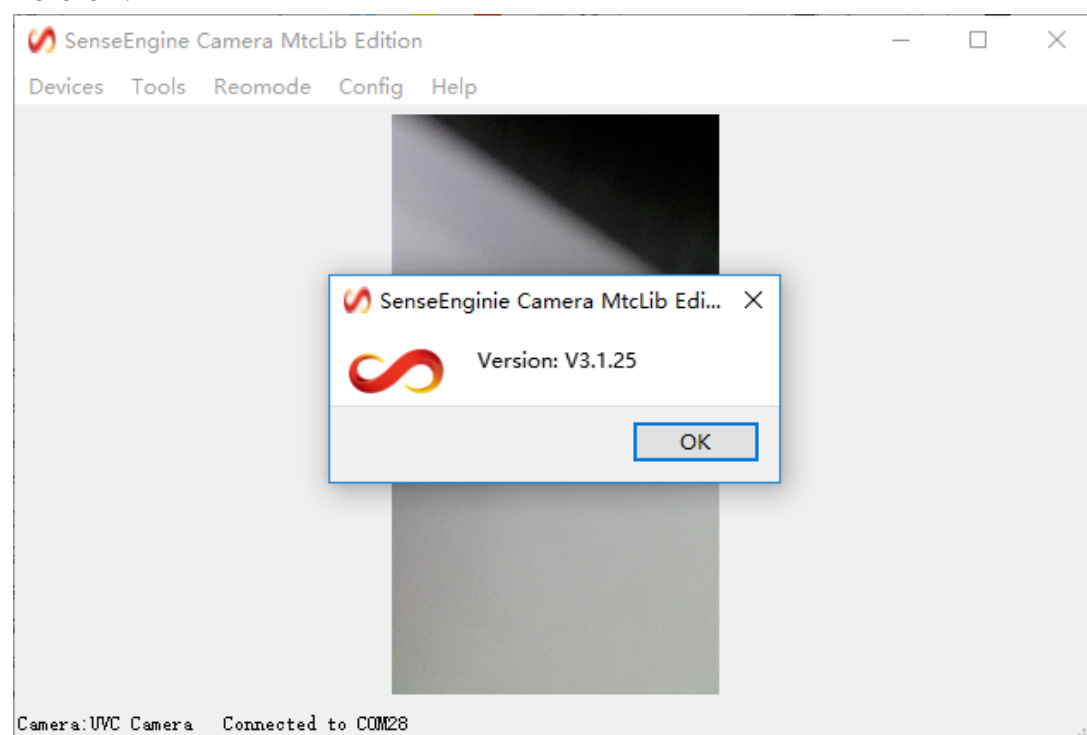
You can select 50hz or 60HZ anti-flash enable or disable in the submenu



33) Software version

Help->About

This button queries the demo software version, not the M30 firmware version.

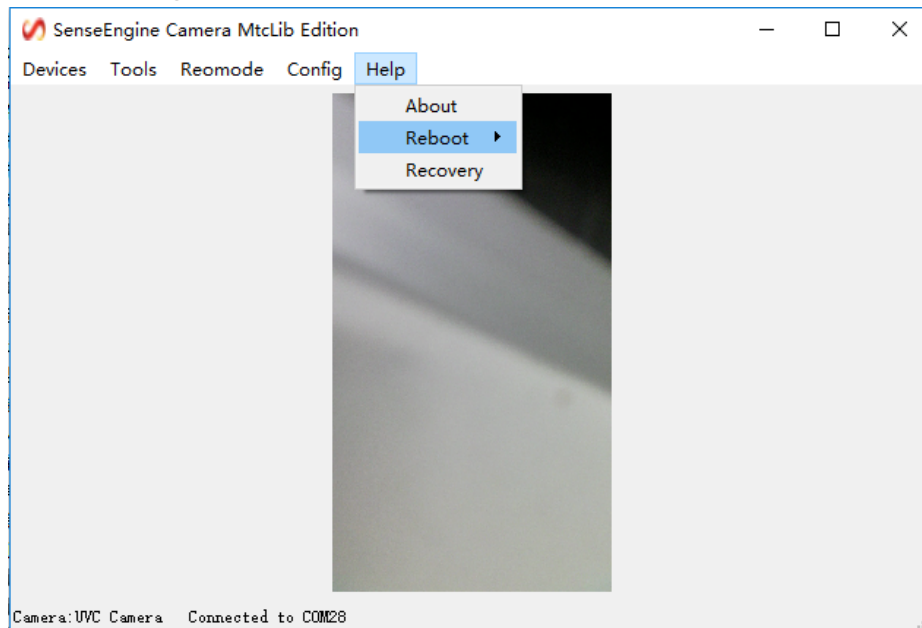


34) Reboot

Help->Reboot

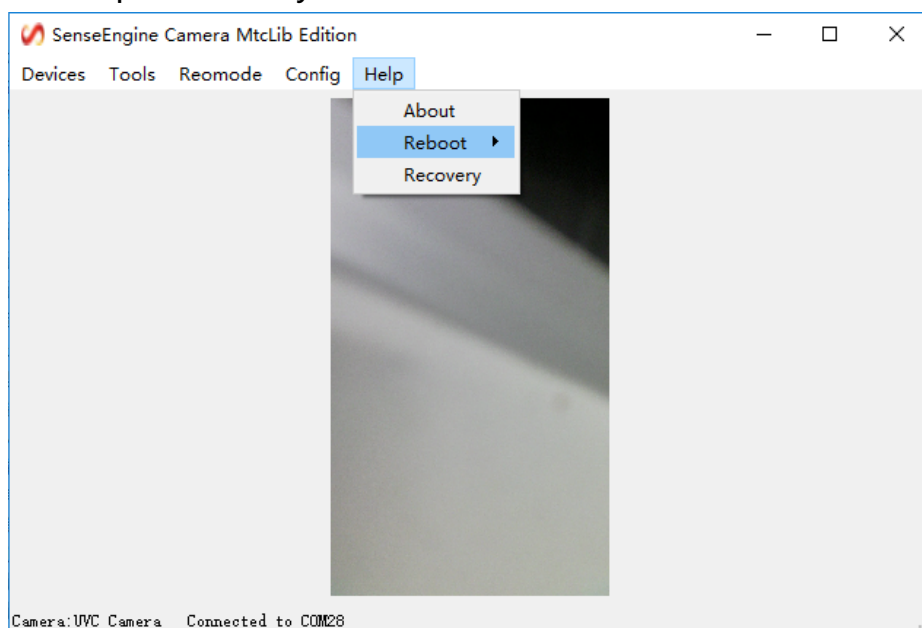
In the submenu, you can select the next startup mode: direct restart, user mode/factory mode, whether it is U disk upgrade mode

Note: After switching to factory mode, it is a virtual network adapter, you can no longer connect with this software; U disk mode will virtual U disk.



35) Reset

Help->Recovery



4.1.2 Ubuntu System

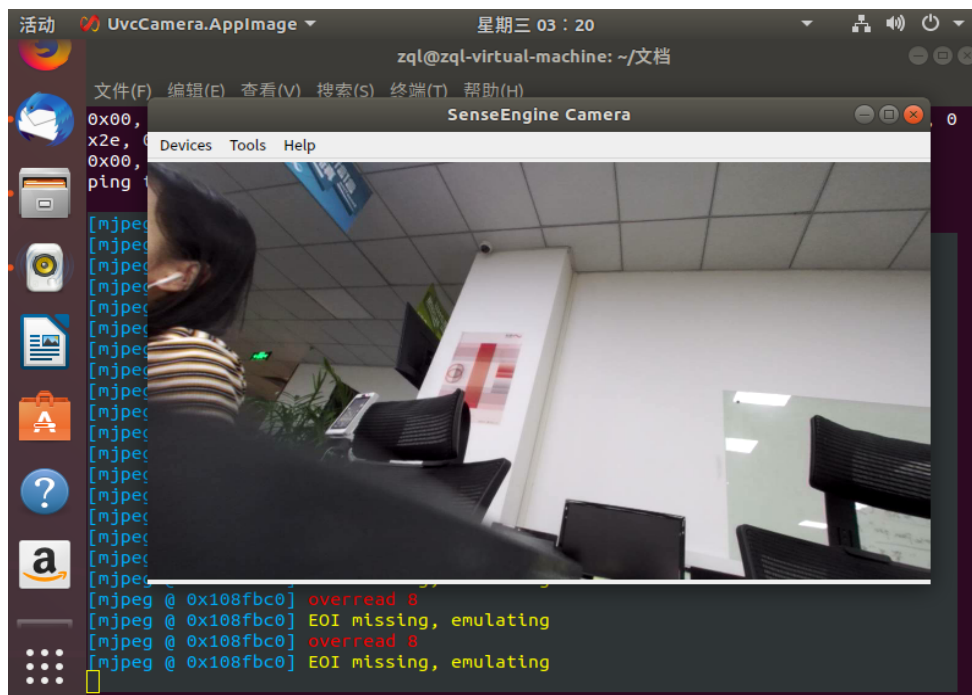
The Linux system currently supports systems higher than Ubuntu14, the software UI is consistent with the windows side, and the functions are the same

1. Connect M30
2. Wait for about 40s to boot up, and the shortcut key Ctrl+Alt+T enters the terminal, enter `cd /dev` to confirm whether the device is connected and the boot is successful, if `ttyACM0` appears, the device is successfully connected. As shown

```
zql@zql-virtual-machine:~$ cd /dev
zql@zql-virtual-machine:/dev$ ls
agpgart      loop3      shm        tty32      tty63      ttyS6
autofs       loop4      snapshot   tty33      tty7        ttyS7
block        loop5      snd        tty34      tty8        ttyS8
bsg          loop6      sr0        tty35      tty9        ttyS9
btrfs-control loop7      stderr     tty36      ttyACM0     uhid
bus          loop-control mapper      tty37      ttyprintk   uinput
cdrom        mcelog     tty        tty38      ttyS0       urandom
cdrw         media0     tty0       tty39      ttyS1       userio
char         mem        tty1       tty4       ttyS10      v4l
console      memory     tty10      tty40      ttyS11      vcs
core         bandwidth tty11      tty41      ttyS12      vcs1
```

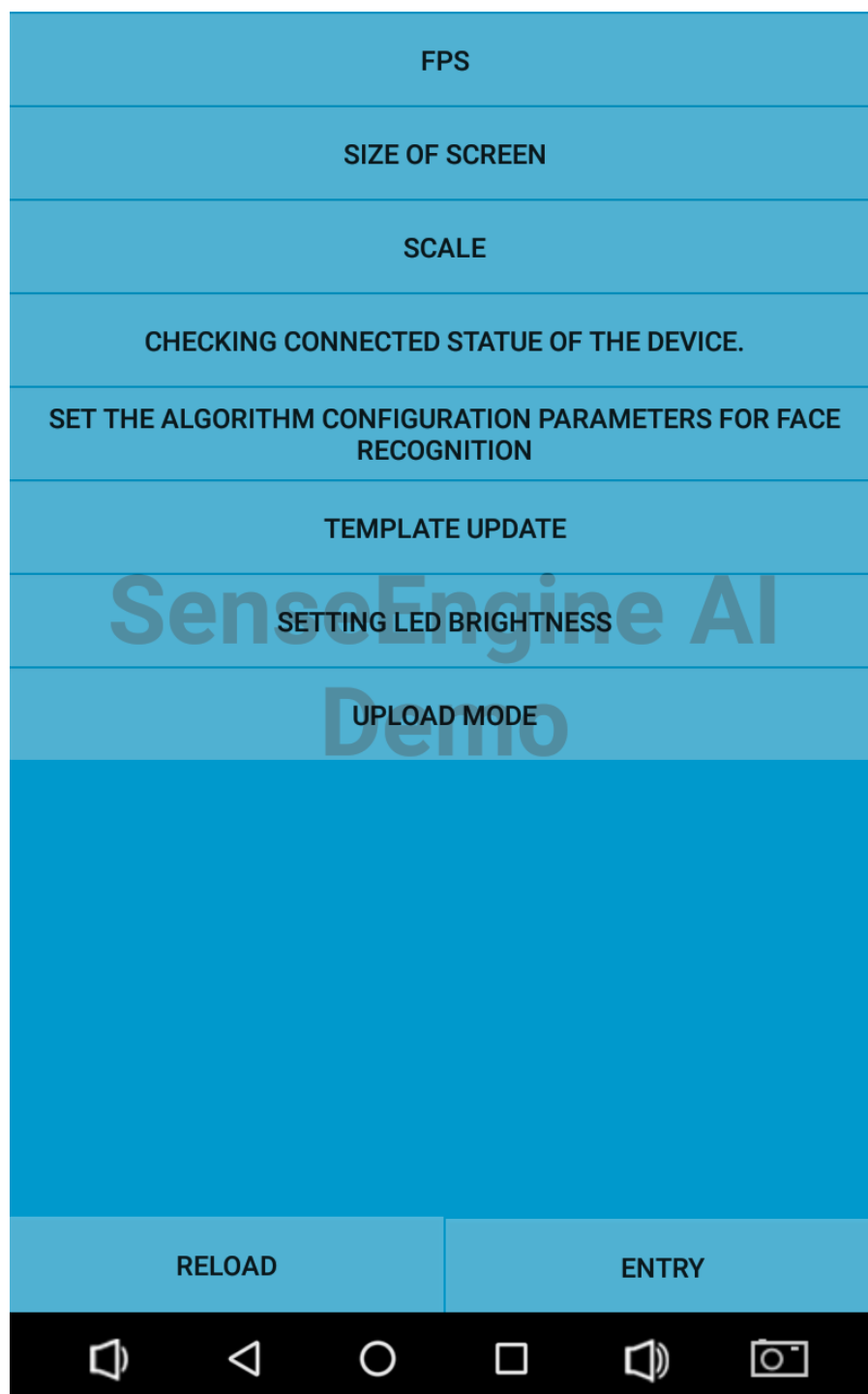
3. Increase permissions
4. Enter the directory to open the application

```
zql@zql-virtual-machine:/dev$ sudo chmod 777 /dev/ttyACM0
zql@zql-virtual-machine:/dev$
```



4.2 Android System

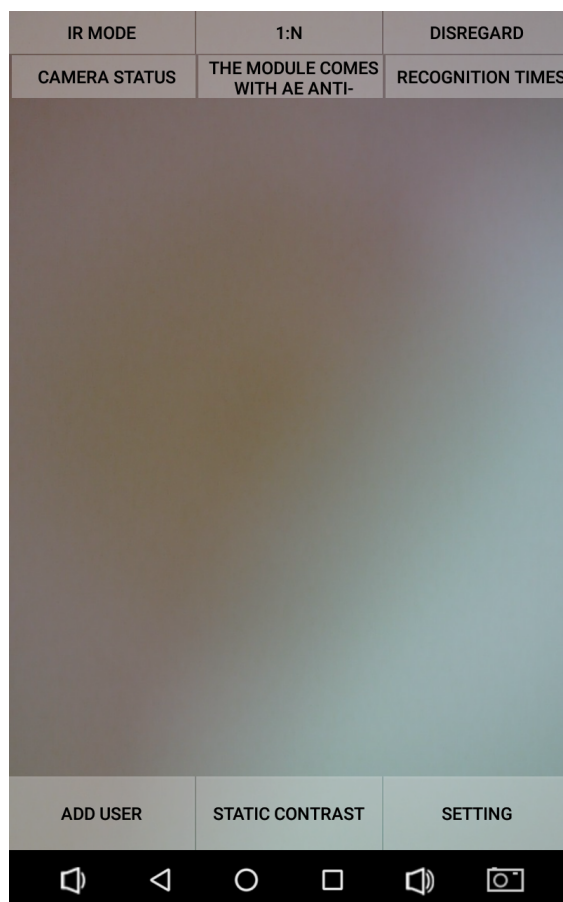
4.2.1 UI interface



4.2.2 Function interface

- 1、 Camera screen rotation: Click to adjust the rotation angle of M30. This interface needs to choose the most suitable angle according to the user's camera placement and display interface.
- 2、 Frame number (FPS): The larger the frame, the smoother the picture. The higher the data processing capacity of the tablet, the range is 10-25 frames per second.
- 3、 Resolution: 360P, 720P, the default is 720P。
- 4、 Horizontal and vertical screen switching: horizontal or vertical screen can be set, according to the user display interface, select horizontal or vertical screen.
- 5、 Stretch: Horizontal stretch or vertical stretch. This is not an interface. The source code is displayed in the demo, which can be selected according to the user's display interface. Developers can also adjust themselves.
- 6、 Check device connectivity: check whether the camera is connected to the Android device.
- 7、 Set the algorithm configuration parameters of face recognition: Refer to the interface document, which is marked with the meaning of each parameter value.
- 8、 Set the IR light brightness value: Set the IR light brightness value and adjust the imaging brightness of the IR camera.
- 9、 Face recognition data reporting mode :
 - ① Actively report, this mode does not need to open the video stream, no need to draw the image, as long as the camera is turned on, the identification data can be sent to the host computer.
 - ② The frame data carries face recognition data. This mode uploads the video stream data with face recognition data at the same time. The video stream must be turned on and the image needs to be drawn.
- 10、 Reload the device: re-initialize and reconnect with the device serial port.
- 11、 Enter: Enter the camera interface.
 - 1) IR/RGB mode switch
 - 2) 1 : N face recognition (Note: The face of the camera is compared with the N faces in the base library for recognition and comparison.)
 - ① :1 :N Single face: Only the largest face is recognized, not whether

it is alive.



2 : 1 : N Multi-face: Recognize multiple faces, the maximum face can be set in the AI algorithm configuration interface, up to five, and does not detect whether it is alive.

3 : 1 : N Single face with live body detection: identify the largest face and detect whether it is alive

4 : 1 : N Multiple faces and live detection enabled: Recognize multiple faces and detect whether they are alive.

- 3) Pause recognition/Resume recognition: Pause face recognition and resume face recognition.
- 4) Camera switch: turn the camera on or off.
- 5) Add users: add users in real time through camera, you can choose whether to store the captured images locally at that time.
- 6) Static comparison: Real-time comparison with people in Curry through the camera, the closest face ID and recognition score will be returned.

7) Settings: Enter the settings page.

DELETE USER	GET ALL IDS
DELETE ALL USER	ADD FEATURES
ADD USER	GET FEATURES
QUERY WHETHER THE BASE LIBRARY ID EXISTS IN THE IDENTIFICATION BASE	DEVICE INFO
UPGRADE FIRMWARE	FACTORY RESET
1:1	BRIGHTNESS
STATIC CONTRAST	GET SN OF EQUIPMENT
QUANTITY OF ID	GET MODEL OF EQUIPMENT
REBOOT SYSTEM	com.sensetime.demo : 3.0.1 Build code:5680



- Delete the specified user: delete the specified user through the ID parameter passed in.
- Clear all users: clear all users in the camera library.
- Adding users: Adding users by passing in ID and picture parameters must be performed in a sub-thread, which takes a long time.
- Query whether the ID exists in the bottom database: By passing in the ID, query whether the user ID is included in the bottom database
- Upgrade firmware: You can update the firmware of the camera module
- 1:1 :Note: The camera face is authenticated and compared with the uploaded face to determine whether it can pass. There are four modes in it, which are the same as 1:N. You can refer to the

1:N mode.

- g) Static comparison: compare the uploaded image with the users in the module base library, and return the closest face ID and score.
- h) Number of bottom library images: Query the number of bottom library images of the module, that is, the number of users.
- i) Reset: reboot system of M30.
- j) Obtain eigenvalues: As long as a user is successfully added, a feature value will be generated in the camera module through the face recognition algorithm. This feature value can be obtained and added to another camera, so that the other camera can also recognize the person's ID without re-importing the picture.
- k) Add eigenvalues : Obtain the eigenvalues and add it to another camera through this interface, so that the other camera can also recognize the person ID without re-importing the picture.
- l) Device information: get the device information of the module camera.
- m) Restore factory settings: clear all data of the module camera.
- n) Get current light sensitivity value: Get the light sensitivity value of the camera.
- o) Get SN: Get the SN of the current module equipment.
- p) Get the device model: currently there are M10, M20, M30.

5 Safe operation and maintenance

5.1 Transport and storage

During transportation, it is necessary to ensure that there is no strong collision and high temperature and high humidity environment. A bad environment will cause severe effects on the equipment to varying degrees, and ultimately affect the use and stability of the product.

5.2 Exception handling mechanism

Abnormal self-restart mechanism: During the use of the device, if the services running in the M30 device stop running due to force majeure or other unknown reasons, the main system daemon will detect and restart the device in time to ensure that all services are working properly.

5.3 Daily maintenance

The upper computer needs to monitor the online status of the equipment. When the equipment is offline or abnormal, the system needs to remind the system according to the preset rules. In the event of an abnormal operation, the device will ensure stable operation of the device by means of automatic restart, manual restart, manual inspection and restoration of network connection, etc.

Front-end equipment maintenance requirements: the use environment is limited to indoor/semi-outdoor conditions, keep the environment relatively dry, the best working temperature is 25 °C, and regularly check whether there is malicious damage and other situations.

Appendix-1. Precautions

- During the use of the equipment, avoid the risk of water entering the equipment itself. Uncontrollable damage will occur after the equipment enters the water, and it may not be possible to retrieve the customer's data in the equipment.
- During the restarting process of the device, avoid plugging or unplugging

the power supply. This will cause uncontrollable system file loss during the restarting process of the device, which will affect the normal use of the device.

Appendix-2. Normal Question FAQ

- (1) The device image has black spots or stains

Answer: Please check whether the glass lens on the front of the camera is stained. If there is stain on the exterior, you can wipe it and use it normally. If it is inside the device, please help after-sales to clean it.

- (2) The imaging side of the device is clear and the other side is blurred

Answer: The camera installation position is not right. It is recommended to check whether the device body is parallel to the upper computer installation plane. If it is not parallel, after-sales assistance is recommended.

Appendix-3. After-sales service

If you have any questions during the use of the product, please contact us through the official after-sales feedback channel:

After-sales mailbox: stas.service@sensetime.com

After-sales hotline: 400-175-5858

Service time: Monday to Friday 9:00-18:00 (except weekends and statutory holidays)

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Business Inquiry

business@sensetime.com

Media Contact

pr@sensetime.com